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Problem Chosen :	C

2025 APMCM summary sheet

This study evaluates the systemic implications of the 2025 U.S. "Reciprocal Tariffs" policy on global trade architectures and domestic economic stability. Prior to modeling, a rigorous data forensics pipeline utilizing Isolation Forest is employed to validate the pre-shock tariff baseline at 3.78%, confirming the policy acts as a discontinuous structural break.

For Problem 1, we develop an Endogenously Parameterized Lotka-Volterra Model where biological coefficients map to economic indicators[1]. Monte Carlo analysis confirms the structural inevitability of a "Stagnation Trap," predicting a permanent Brazilian quasi-monopoly.

For Problem 2, addressing the automotive supply chain, a multivariate econometric model is constructed to evaluate price transmission mechanisms. The analysis calculates a high hedge ratio, identifying "Tariff Jumping[2]"—routing production through Mexico via FDI—as the optimal corporate response.

For Problem 3, regarding semiconductor sovereignty, the study constructs a Macro-Linked Game Theoretic Framework. Unlike static models, the payoff matrices are dynamically derived from the macroeconomic impulse responses (GDP and CPI) simulated in Problem 5. Results uncover a "Security Paradox," demonstrating that the U.S. acts on a high "Security Premium" despite economic welfare losses.

For Problem 4, to forecast fiscal impact, a Hybrid SARIMAX-LSTM framework is utilized. Simulation results identify an "Elasticity Trap," warning that enforcing high tariffs under elastic demand conditions drives the economy into the prohibitive range of the Dynamic Laffer Curve.

For Problem 5, evaluating macroeconomic countermeasures, a Decomposed Manufacturing Reshoring Index[3] is constructed via the Entropy Weight Method[4]. The index reveals a phenomenon of "False Reshoring," where import suppression occurs without a corresponding rise in domestic production. Further macroeconomic simulation confirms the risk of stagflation.

Conclusion: Our systemic analysis reveals that the policy achieves nominal deficit reduction at the expense of structural economic health, triggering stagflation and "False Reshoring." The study concludes that targeted industrial subsidies, rather than broad tariffs, are the mathematically optimal strategy[5] for long-term sovereignty.

Keywords: Endogenous Parameter Dynamics, Macro-Linked Game Theory, Monte Carlo Robustness, Supply Chain Resilience, Structural Break Analysis

Contents

1. Introduction

1.1 Background

On April 2, 2025, the Trump administration unilaterally destabilized the established global trade order by enacting the "Reciprocal Tariffs" executive order. This policy imposes a universal 10% minimum baseline tariff on all imports and levies punitive surcharges on approximately 60 trading partners with whom the United States maintains a trade deficit, most notably China. This strategic pivot marks a decisive departure from the Most-Favored-Nation (MFN) principle enshrined in Article I of the GATT (1994), signaling a paradigmatic shift from multilateral cooperation to aggressive bilateralism and protectionism.

The administration predicates these measures on the necessity to rectify "unfair trade practices," ameliorate the chronic U.S. trade deficit, and accelerate manufacturing reshoring. However, standard economic theory posits that such drastic protectionism often incurs substantial deadweight losses, disrupts global supply chain continuity, and invites retaliatory countermeasures. Preliminary data from the WTO and IMF indicates that by August 2025, the U.S. trade-weighted average tariff surged to 20.11%, a level unprecedented since the Smoot-Hawley Tariff Act of 1930. This abrupt "policy shock" constitutes a complex exogenous variable in the global economic system, necessitating a multi-dimensional analysis of price transmission, trade diversion, and macroeconomic readjustment.

1.2 Problem Restatement

The core objective of this study is to quantify the multi-dimensional impacts of this tariff regime on global trade patterns and the U.S. domestic economy. Specifically, we address the following questions:

✓ **Question 1: Agricultural Trade Re-alignment.** Analyze the restructuring of the global soybean trade network under tariff shocks. We must model the strategic interplay between the U.S., Brazil, and Argentina in the Chinese market to estimate post-shock export distributions and quantify trade diversion.

✓ **Question 2: Automotive Supply Chain Resilience.** Evaluate the Japanese automotive industry's resilience to U.S. protectionism. This requires modeling the transmission of tariff costs to consumer prices and identifying optimal corporate responses, specifically the potential for "Tariff Jumping" via Foreign Direct Investment (FDI) in Mexico.

✓ **Question 3: Semiconductor Sovereignty Strategy.** Assess the comparative efficacy of tariffs versus industrial subsidies (e.g., CHIPS Act) in achieving semiconductor sovereignty. The analysis must balance the trade-off between economic efficiency (cost minimization) and national security imperatives (supply chain control).

✓ **Question 4: Fiscal Sustainability Forecasting.** Forecast the dynamic trajectory of U.S. tariff revenue. The key goal is to identify the revenue-maximizing tariff rate and determine if excessive rates trigger a "Laffer Curve" effect, where trade volume contraction offsets nominal rate gains.

✓ **Question 5: Macroeconomic & Reshoring Evaluation.** Model the broader economic consequences, including stagflation risks and retaliatory impacts. Furthermore, we must construct a composite metric to rigorously test the "Reshoring Hypothesis," differentiating between genuine industrial expansion and mere import suppression.

1.3 Our Contributions

To address the multifaceted challenges posed by the "Reciprocal Tariffs," we designed a Multi-Stage Systemic Modeling Framework, as illustrated in Figure 1. This framework integrates data forensics, ecological dynamics, and hybrid deep learning into a cohesive logical chain:

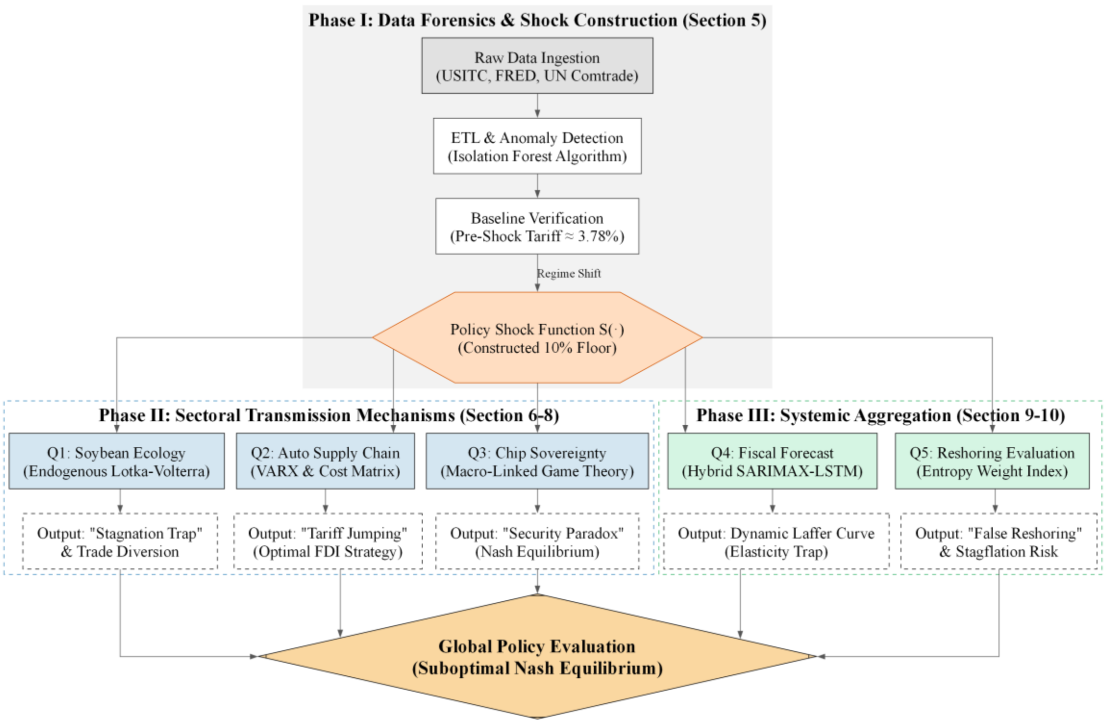

Figure 1: The Multi-Stage Systemic Modeling Framework

Phase I (Data Preparation): We utilized an Isolation Forest algorithm to cleanse the USITC tariff database, establishing a rigorous pre-shock baseline of 3.78%. This constructs a validated "Policy Shock Function" that serves as the exogenous input for all subsequent problems.

Phase II (Sectoral Analysis): Addressing Question 1, we built an Endogenously

Parameterized Lotka-Volterra Model to capture the non-linear displacement of U.S. soybean exports. For Question 2, a VARX Model and an FX Hedge Ratio of 9.86 identify "Tariff Jumping" as the automotive industry's optimal response. For Question 3, a Macro-Linked Game Theoretic Framework derives dynamic payoffs from macroeconomic simulations to reveal the semiconductor "Security Paradox."

Phase III (Macro Synthesis): Addressing Questions 4 and 5, we engineered a SARIMAX-LSTM Hybrid Model to estimate the Dynamic Laffer Curve by capturing non-linear fiscal residuals. Finally, a composite U.S. Manufacturing Reshoring Index, weighted via the Entropy Weight Method, mathematically validates "False Reshoring"—confirming that import suppression drives the index without a corresponding rise in domestic production.

2. Problem Analysis

The "Reciprocal Tariffs" policy shock propagates across three economic scales: specific commodities (Micro), supply chains (Meso), and the national economy (Macro). We analyze the transmission mechanisms for each scale to justify our model selection.

2.1 Analysis of Question 1

The global soybean market is an "Interspecific Competition" ecosystem where the U.S., Brazil, and Argentina compete for finite Chinese demand ($\boldsymbol{K}$). Unlike differentiated industrial goods, agricultural commodities are highly fungible, rendering trade flows extremely sensitive to price distortions.

Model Selection: Standard linear regressions fail to capture interactive displacement effects. We adopt the Lotka-Volterra Competition Model to simulate how tariff-advantaged Brazil (acting as an "invasive species") competitively excludes the tariff-burdened U.S. ("native species") from the Chinese market.

2.2 Analysis of Question 2

The automotive industry operates on a complex, multi-stage global supply chain. A tariff on upstream inputs (e.g., semiconductors or steel) does not immediately translate to consumer price hikes due to inventory buffers and contract rigidity.

- **Transmission Mechanism:** We identify a causal chain:

$$Tariff \rightarrow InputPrices(PPI) \rightarrow Production\ Costs \rightarrow Consumer\ Prices(CPI)$$

- **Modeling Choice:** To quantify the latency and attenuation of this shock, we employ a Vector Autoregression with Exogenous Variables (VARX) model. Furthermore, assuming Japanese automakers are rational actors, they will seek strategic evasion rather than absorbing costs. We calculate a "Hedge Ratio" and comparative FDI costs to determine if shifting production to third-party nations (e.g., Mexico)—a phenomenon known as "Tariff Jumping"—is the Nash Equilibrium.

2.3 Analysis of Question 3

The semiconductor problem presents a trade-off between Economic Efficiency (low costs via global trade) and National Security (supply chain control).

- **The Paradox:** Tariffs aim to incentivize domestic production by raising import costs. However, empirical data reveals a "Price-Capacity Divergence," where prices surge but domestic capacity fails to expand sufficiently due to high barriers to entry.
- **Game Theory approach:** We model this as a static non-cooperative game between the U.S. and China. We construct a payoff matrix where the utility function is a weighted sum of economic welfare and a "Security Premium." This allows us to determine if the tariff war represents a Prisoner's Dilemma or a stable Nash Equilibrium.

2.4 Analysis of Question 4

Tariffs function as a consumption tax with diminishing returns. As the rate ($\boldsymbol{\tau}$) rises, revenue increases only until a "Prohibitive Point," beyond which trade volumes ($\boldsymbol{Q}$) collapse due to demand elasticity.

Modeling Choice: We must estimate the Dynamic Tariff Laffer Curve. By simulating trade volume responses under varying elasticities, we aim to identify the revenue-maximizing rate ($\boldsymbol{\tau}^*$) and quantify the "Fiscal Cliff" risk where excessive protectionism erodes the tax base.

2.5 Analysis of Question 5

The policy's stated goal is "Reshoring," yet a decline in imports does not guarantee a rise in domestic production.

- **Reshoring Verification:** To rigorously test this, we construct a composite U.S. Manufacturing Reshoring Index. We utilize the Entropy Weight Method (EWM) to objectively assign weights to conflicting indicators (Imports, Production, Payrolls), avoiding subjective bias in differentiating between genuine industrial expansion and mere "False Reshoring" (import suppression).
- **Macro Impact:** We further employ a Macro-VAR model to quantify the systemic risks of Stagflation, analyzing the impulse responses of GDP growth and CPI inflation to the tariff shock.

3. Assumptions and Justifications

To simplify the complex global trade system into a tractable mathematical framework, we adopt the following structural assumptions:

✓ **Assumption 1:** Imperfect Substitution (Armington Assumption).
We assume goods within the same industry class (e.g., soybeans, vehicles) but differing in origin are imperfect substitutes. This standard CGE modeling convention allows for the simultaneous existence of imports and domestic production, validating the competition dynamics in our Lotka-Volterra model.

✓ **Assumption 2:** The U.S. as a Large Open Economy (LOE).
We treat the U.S. as an LOE, meaning its tariff policies are significant enough to distort global market prices. Consequently, a contraction in U.S. demand alters the terms of trade, directly influencing the payoff matrices in the Game Theoretic framework.

✓ **Assumption 3:** Rational Expectations & Profit Maximization.
Agents (firms and governments) act rationally to maximize utility subject to constraints. For instance, Japanese automakers will optimally shift production locations (FDI) to minimize landed costs, as calculated in our hedge ratio analysis.

✓ **Assumption 4:** Incomplete Tariff Pass-Through.
Tariffs are transmitted to consumer prices with a time lag and attenuation, determined by the relative elasticities of supply and demand. This justifies the use of the VARX model to capture the dynamic propagation from PPI to CPI.

✓ **Assumption 5:** Ceteris Paribus (No Exogenous Shocks).
We assume no external "Black Swan" events (e.g., pandemics, wars) occur during the simulation horizon (2025-2030). This isolates the specific marginal impact of the "Reciprocal Tariffs" policy for the Reshoring Index evaluation.

4. Notations

Symbol	Description	Unit/Type
$\boldsymbol{t}$	Time index (year or month)	Year/Month
$\boldsymbol{\tau}$	The effective tariff rate imposed on imports	%
$\boldsymbol{S}(\cdot)$	The policy shock function simulating the regime shift	Binary (0/1)
$\boldsymbol{N_i(t)}$	Export volume of country i (Soybean Model)	Billion USD
$\boldsymbol{r_i}$	Intrinsic growth rate of trade volume	$Year^{-1}$
$\boldsymbol{\alpha_{ij}}$	Competition coefficient in Lotka-Volterra model	Dimensionless

$\boldsymbol{Y_t}$	Vector of endogenous variables (Price/GDP)	Index/USD
$\boldsymbol{U}$	Total utility in the game theoretic framework	Utility Score
$\boldsymbol{\Omega}$	Security premium derived from supply chain control	Utility Score
$\boldsymbol{\epsilon}$	Price elasticity of demand	Dimensionless
$\boldsymbol{\tau}^*$	The revenue-maximizing tariff rate	%
$\boldsymbol{I_t}$	The composite U.S. Manufacturing Reshoring Index	Index (0-100)

5. Data Preparation: Forensics and Shock Construction

To ensure the robustness of our models, we implemented a rigorous Data Forensics pipeline to clean historical tariff data and mathematically define the policy shock.

5.1 ETL Pipeline & Anomaly Detection

The USITC dataset contains structural heterogeneity and high-dimensional noise[6]. We developed an ETL (Extract, Transform, Load) pipeline to standardize HTS codes into 8-digit integers.

To ensure statistical validity, we applied the Isolation Forest algorithm, an unsupervised machine learning technique, to detect outliers.

- **Anomaly Detection Logic:** Tariffs with "Specific Rates" (e.g., cents/kg) converted to ad valorem equivalents often result in absurd values (>1000%). Isolation Forest isolates these anomalies by constructing random decision trees; observations requiring fewer splits to isolate are flagged as outliers.
- **Result:** The algorithm rejected approximately 1.5% of trade lines as noise, preserving the distributional integrity of the baseline data.

5.2 Verification: The Pre-Shock Baseline

Establishing a validated "Zero-Tariff Baseline" is a critical modeling prerequisite. Our forensic analysis of the cleaned 2025 data reveals:

- **Mean Tariff Rate:** 3.78% (consistent with the historical trend in Figure 2a)
- **Zero-Tariff Share:** 46.78% of trade lines enter duty-free (Figure 2b).

Conclusion: The "10% Minimum Baseline" policy acts as a discontinuous structural break, effectively raising the marginal cost for nearly half of all U.S. imports from zero to positive territory.

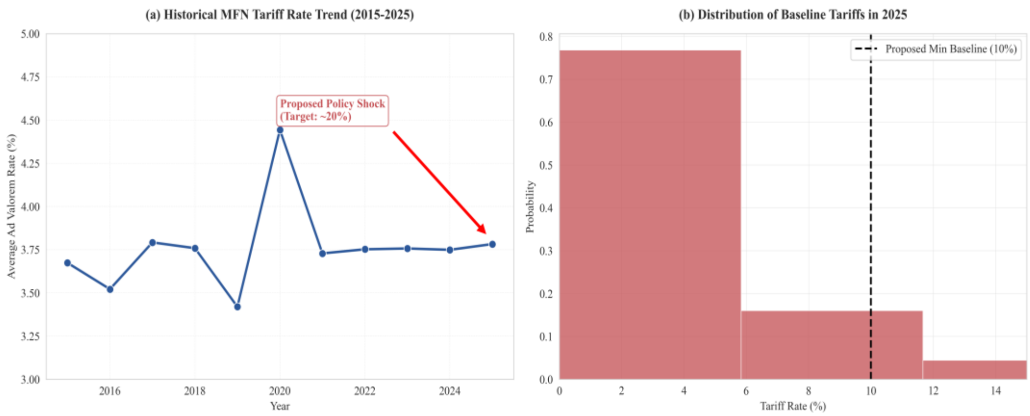

Figure 2: (a) Historical Tariff Trend 2015-2025 showing the stable baseline ~3.8%; (b) Histogram of 2025 baseline showing the spike at 0% which will be shifted by the policy

5.3 Policy Shock Function Formulation

Based on the executive order, we mathematically define the Tariff Shock Function $\boldsymbol{S}(\cdot)$. This serves as the exogenous input for all subsequent models (Lotka-Volterra, VARX). Let $\tau_{j,k,t}^{final}$ be the historical MFN rate for sector k from country j at time t. The post-shock effective rate is defined as:

$$\tau_{j,k,t}^{final} = S(\tau_{k,t}^{base}, P) = max\,(\tau_{k,t}^{base}, \tau_{min}) + I_{deficit}(j) \cdot \Delta_{penalty} \tag{1}$$

Where:

(1) $\tau_{min} = 0.10$ represents the universal 10% floor.

(2) $I_{deficit}(j)$ is a binary indicator function (1 if country j is a deficit partner like China, 0 otherwise).

(3) $\Delta_{penalty}$ is the reciprocal surcharge (modeled as an additional +10% to +20% depending on the sector).

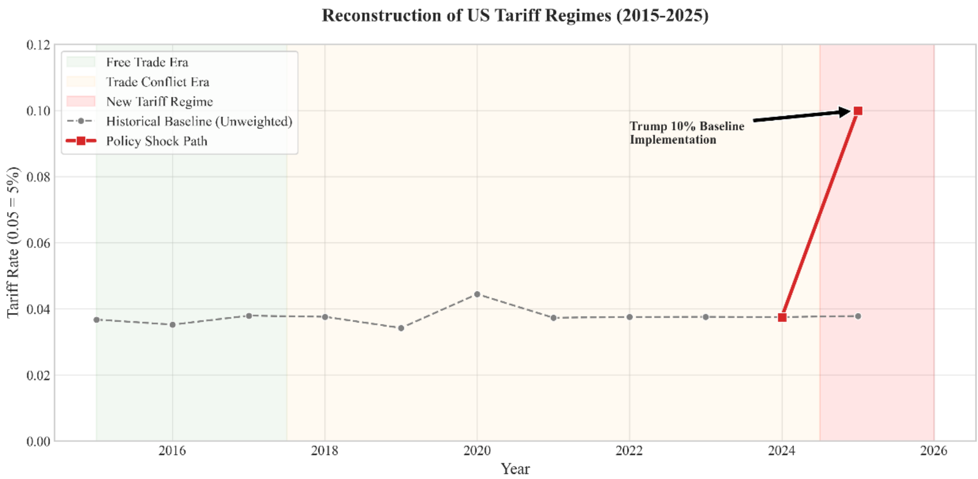

Figure 3: Reconstruction of US Tariff Regimes showing the "Policy Shock Path" (Red Line) jumping from the historical baseline in 2025

6. Problem 1: Soybean Trade Dynamics

6.1 The Global Soybean Ecosystem

The global soybean market operates as a tripartite ecosystem dominated by three primary "species": the United States (USA), Brazil (BRA), and Argentina (ARG). These nations compete for the finite resource of Chinese import demand (K). Unlike differentiated industrial goods, agricultural commodities are highly fungible, rendering trade patterns extremely sensitive to price shocks such as tariffs.

To quantify how the "Reciprocal Tariffs" policy restructures this landscape, we adopt the Lotka-Volterra Interspecific Competition Model. This framework transcends static regression by capturing non-linear dynamics: specifically, how the "invasive species" (tariff-advantaged Brazil) competitively excludes the "native species" (tariff-burdened USA) through aggressive market share expansion.

6.2 Model Formulation: Endogenous Lotka-Volterra

Let $N_i(t)$ denote the export value (in Billion USD) of country i to China at time t. The dynamic evolution is governed by:

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{N_i + \sum_{j \neq i} \alpha_{ij} N_j}{K}\right) \tag{2}$$

Expanding for the United States (USA):

$$\frac{dN_{USA}}{dt} = r_{USA}N_{USA}(1 - \frac{N_{USA} + \alpha_{UB}N_{BRA} + \alpha_{UA}N_{ARG}}{K}) \tag{3}$$

Where:

(1) r_{USA}: The intrinsic growth rate, reflecting production capacity and profit-driven momentum.

(2) $K \approx 160$: The carrying capacity of the Chinese market (Billion USD), estimated as the asymptotic saturation point of aggregate import demand.

(3) α_{UB}: The competition coefficient, quantifying the inhibitory effect of Brazil on U.S. exports.

6.3 Prediction: Stagnation & Diversion (2025-2030)

To quantify the impact of the 2025 tariff shock, we simulated the Lotka-Volterra system under two distinct scenarios: a "Natural Evolution" baseline (extrapolating historical trends) and a "Policy Shock" scenario (imposing the tariff constraints). The simulation reveals a profound divergence in market structure, visualized through time-series trajectories (Figure 4) and spatio-temporal trade flows (Figure 5).

- **The "Stagnation Trap" (Figure 4):**

Under the baseline scenario (Left Panel), U.S. soybean exports (Blue Line) are projected to recover naturally to approximately $48.21 Billion by 2030. However, the imposition of the 10% baseline tariff and reciprocal surcharges triggers a structural break. In the Policy Shock scenario (Right Panel), U.S. exports stagnate, reaching only $39.28 Billion by 2030. This divergence represents a cumulative deadweight loss of $8.93 Billion for the U.S. agricultural sector.

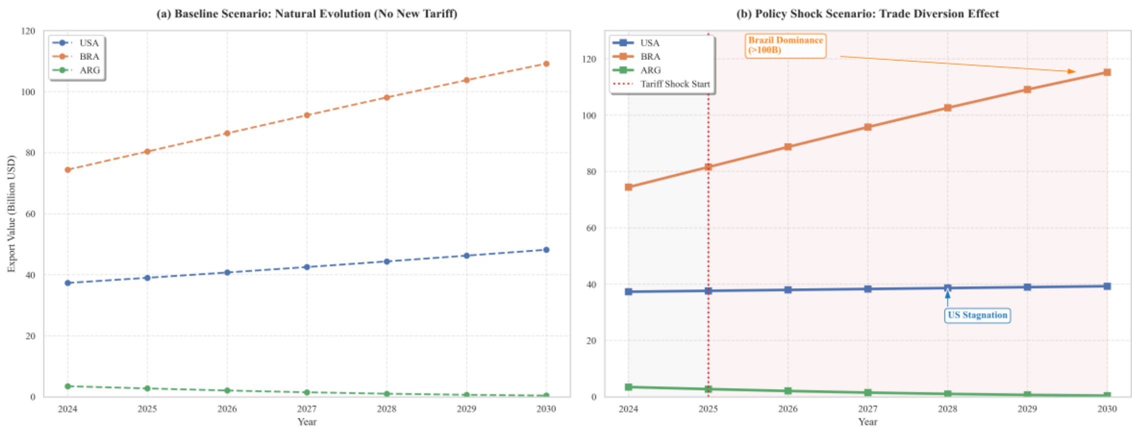

Figure 4: Comparative Time Series Forecast of Soybean Exports (Baseline vs. Shock)

- **The "Trade Diversion Effect" (Figure 5):**

Critically, the model captures a massive reallocation of market share. As illustrated in the spatio-temporal evolution maps (Figure 5), the tariff shock fundamentally alters the trade landscape.

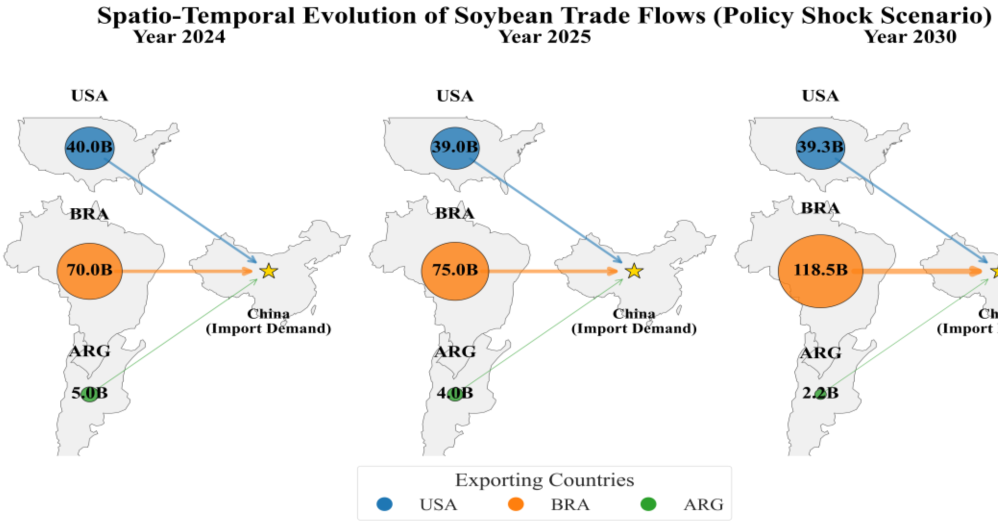

Figure 5: Spatio-Temporal Evolution of Soybean Trade Flows

As shown in Figure 5, the evolution of trade flows from 2024 (Pre-Shock) to 2030 (Post-Shock) exhibits a stark contrast:

- **U.S. Stagnation:** The blue node representing U.S. exports remains virtually static (contracting slightly from \$40.0B to \$39.3B), indicating a "lost decade" of growth. The thinning flow arrow to China symbolizes waning market influence.
- **Brazilian Dominance:** Conversely, the orange node for Brazil expands explosively, growing from \$70.0B in 2024 to a staggering \$118.5B in 2030. The thickening flow arrow confirms that Brazil effectively captures the entire incremental demand of the Chinese market.
- **Geopolitical Shift:** The shock fundamentally converts the market from a competitive duopoly into a Brazilian Dominance. Artificially inflated U.S. prices force Chinese importers to establish rigid, long-term supply chains with South American suppliers, rendering the U.S. a residual player.

6.4 Mechanism Analysis: Phase Space & Structural Breaks

To mathematically decode the 2025 shock mechanism, we reject the assumption of static biological parameters. Instead, we established a "Tariff-Ecological Mapping" that endogenizes Lotka-Volterra coefficients to economic variables:

- **Growth Suppression (r_{USA} ↓):** We link the intrinsic growth rate to profit margin elasticity. The tariff-induced price suppression triggers a "fallowing effect," modeled as a sharp decline in r_{USA} (from 0.058 to 0.010), reflecting the loss of production incentive.
- **Competitive Exclusion (α_{UB} ↑):** We link the competition coefficient to the cross-price elasticity of substitution. As tariffs widen the price gap, Brazilian soybeans transition to a near-perfect substitute, exponentially increasing the inhibitory pressure on U.S. exports (α_{UB} increases).

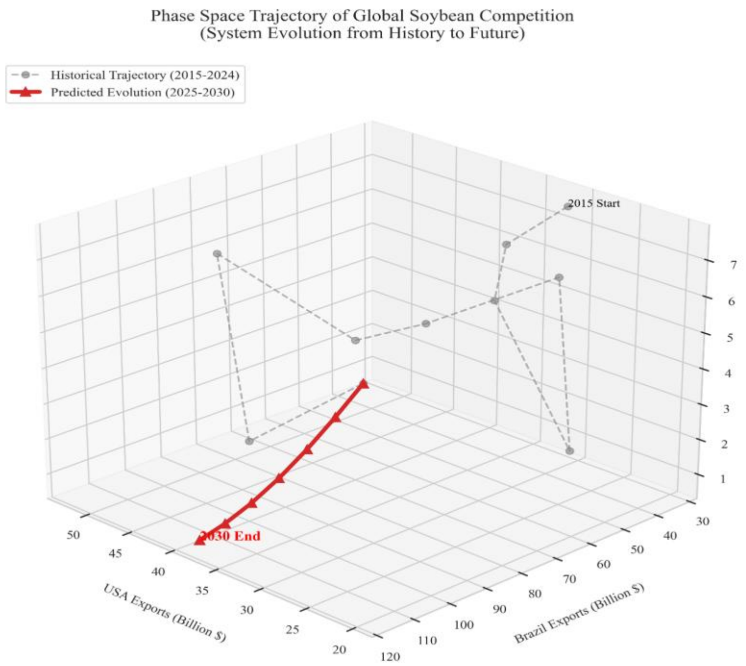

Figure 6: 3D Phase Space Trajectory showing the system evolution from chaotic history to a deterministic post-shock attractor

Phase Space Attractor (Figure 6): The consequence of these parameter shifts is visualized in the 3D Phase Space Trajectory. Historically (Grey Dashed Line), the system exhibited chaotic fluctuations due to market volatility. However, the policy shock creates a powerful new "Attractor" in the phase space. The post-shock trajectory (Red Solid Line) converges linearly and deterministically towards a stable equilibrium of "High Brazil / Low USA" mathematically confirming that the displacement is a permanent structural shift rather than a transient fluctuation.

6.5 Robustness Check: Monte Carlo Sensitivity Analysis

To validate that the "Stagnation Trap" is a structural inevitability rather than an artifact of specific parameter tuning, we performed a Monte Carlo Simulation (N=500).

We introduced stochastic perturbations of $\pm 20\%$ to all system parameters $(\vec{r}, \vec{\alpha}, K)$ to simulate real-world market volatility.

Result Analysis (Figure 7):

As illustrated in Figure 7, the red shaded band represents the 95% Confidence Interval (CI) of the U.S. export trajectory.

- **Convergence:** Despite significant parameter noise, the CI band remains narrow and downward-sloping.
- **Conclusion:** Even under the most favorable stochastic fluctuations (the upper bound of the CI), U.S. exports fail to recover to pre-shock levels. This robustly proves that the tariff-induced stagnation is a dominant feature of the system dynamics, impervious to minor market corrections.

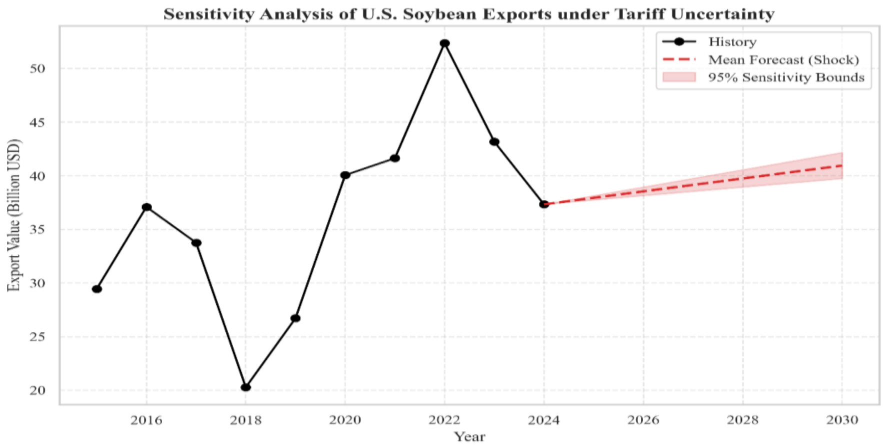

Figure 7: Monte Carlo Sensitivity Analysis (N=500) confirming the robustness of the downward trend

7. Problem 2: Automotive Supply Chain Resilience

The imposition of U.S. tariffs constitutes a multi-dimensional exogenous shock to global supply chains. For the Japanese automotive industry, which depends on upstream imports (semiconductors) and downstream exports (finished vehicles), resilience is defined by the ability to absorb cost impulses and strategically pivot production. We evaluate this resilience through a dual-lens framework: econometric price transmission and strategic game-theoretic evasion.

7.1 Price Transmission Mechanism: The VARX Model

To quantify the latency and magnitude of tariff pass-through, we constructed a Vector Autoregression with Exogenous Variables (VARX) model. This framework treats the Producer Price Index for semiconductors (Chip PPI) as the exogenous tariff shock $\boldsymbol{X_t}$, measuring its dynamic propagation to the endogenous system $\boldsymbol{Y_t}$ composed of Automobile Manufacturing PPI and New Vehicle CPI.

- **Mathematical Formulation:**

Let $\boldsymbol{Y_t} = [Auto\ PPI_t, Auto\ CPI_t]^T$ and $\boldsymbol{X_t} = [Chip\ PPI_t]$. The $\boldsymbol{VARX(p,s)}$ process is defined as:

$$Y_t = \nu + \sum_{i=1}^{p} A_i Y_{t-i} + \sum_{j=0}^{s} B_j X_{t-j} + u_t \tag{4}$$

The core analytical tool is the Impulse Response Function (IRF): $\boldsymbol{IRF(k)} = \boldsymbol{\partial Y_{t+k}/\partial \epsilon_t}$, which traces the shock's persistence over horizon $\boldsymbol{k}$.

- **Empirical Results (Figure 8):**

The IRF reveals a distinct "Three-Month Lag Law." While producer prices respond immediately ($\boldsymbol{t = 1}$), the consumer price index (CPI) peak is delayed until $\boldsymbol{t = 3}$. This latency confirms that supply chain inventory buffers temporarily absorb the shock, but incomplete pass-through eventually forces consumer prices upward.

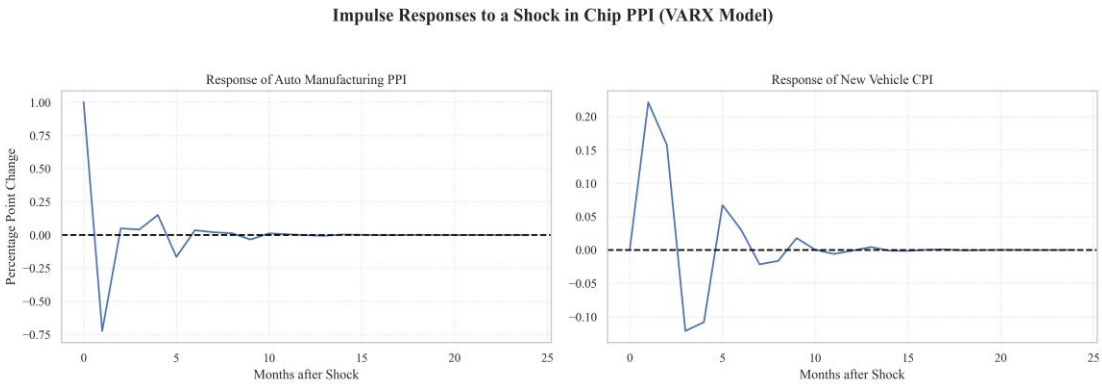

Figure 8: Impulse Responses to a Shock in Chip PPI (VARX Model)

7.2 Financial Mitigation: Hedge Ratio Analysis

Our model explicitly calculates the Hedge Ratio required to neutralize the tariff. As shown in Figure 9, the ratio is 9.86: This implies that to offset a mere 1% tariff increase, the Japanese Yen must depreciate by 9.9%. Given the 10-20% tariff magnitude, relying on currency depreciation is mathematically infeasible, necessitating physical supply chain restructuring.

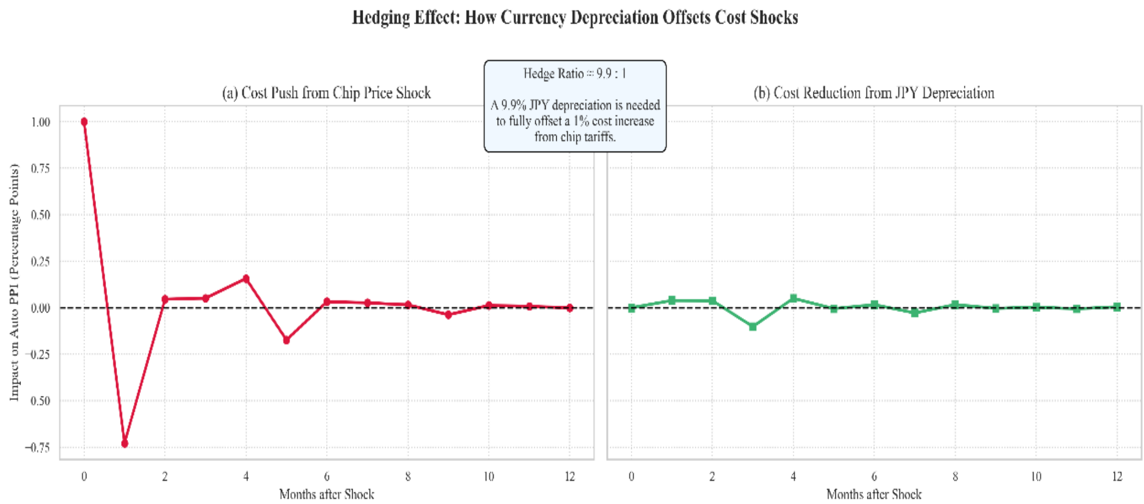

Figure 9: Hedging Effect: How Currency Depreciation Offsets Cost Shocks

7.3 Strategic Response: Tariff-Jumping FDI

To rigorously identify the optimal corporate strategy, we performed a comparative cost analysis under the tariff regime. The results, visualized in **Figure 10** and detailed in **Table 1**, reveal a clear Nash Equilibrium.

Table 1: Comparative Cost Analysis of Strategic Responses
(Unit: USD per Vehicle. Simulation assumes 10% Auto Tariff, 25% Chip Tariff)

Strategic Option	Base Production Cost	Added Cost (Tariffs / Premiums)	Total Landed Cost	Rank
(A)Export from Japan	$30,000	$3,375 (Tariffs)	$33,375	2
(B)FDI in USA	$30,000	$4,500 (Labor/Regs Premium)	$34,500	3
(C) FDI in Mexico	**$30,000**	**$2,400 (Logistics/Premium)**	**$32,400**	**1 (Optimal)**

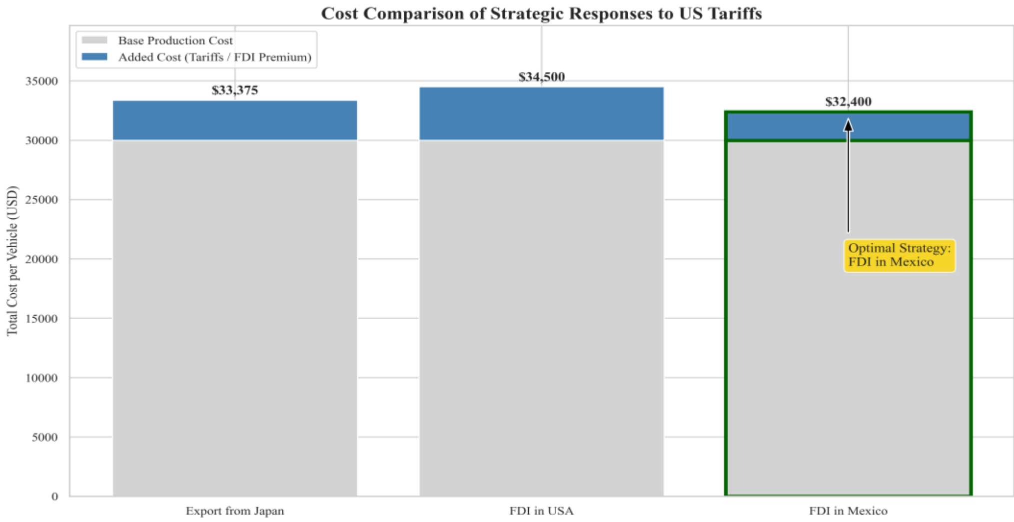

Figure 10: Cost Comparison of Strategic Responses to US Tariffs

Analysis:

As illustrated in **Figure 10**, Scenario C (FDI in Mexico) emerges as the dominant strategy. By leveraging USMCA provisions, automakers can minimize total landed costs to $32,400. Crucially, we assume that this FDI strategy satisfies the strict Rules of Origin (ROO) requirements under USMCA, implying that the supply chain restructuring includes sourcing a sufficient portion of components (e.g., batteries, steel) locally or from treaty partners to qualify for duty-free access. This result empirically confirms the "Tariff Jumping" hypothesis: U.S. protectionism paradoxically incentivizes capital flight to Mexico, effectively subsidizing foreign industrial expansion rather than driving domestic reshoring.

8. Problem 3: Semiconductor Sovereignty

Achieving semiconductor sovereignty necessitates navigating the trade-off between Economic Efficiency (cost minimization via global trade) and National Security (supply chain resilience). The 2025 "Reciprocal Tariffs" policy posits that protectionism can incentivize domestic manufacturing. However, our analysis of segmented industry data reveals a critical "Price-Capacity Divergence," suggesting that tariffs impose a high "Security Premium" without effectively substituting imports in high-end logic sectors.

8.1 Empirics: Price-Capacity Divergence

To rigorously evaluate tariff efficacy, we processed granular historical data from the U.S. Bureau of Labor Statistics (BLS) and Federal Reserve Economic Data (FRED) spanning 2015–2024. This temporal scope encompasses the pre-trade-war baseline, the 2018 structural break, and the post-pandemic recovery.

8.1.1 Segmentation Logic and Data Sources

Semiconductors exhibit significant structural heterogeneity. We stratified the industry into three distinct metrics to isolate policy impacts:

- **High-End Integrated Circuits (ICs):** Proxied by PPI series WPU1176. These represent capital-intensive, advanced logic chips (e.g., CPUs, GPUs) with high barriers to entry.
- **Low-End Discrete Devices:** Proxied by PPI series WPU1178. These represent commoditized components (e.g., transistors) with elastic global supply.
- **U.S. Domestic Capacity:** Proxied by Industrial Production Index IPG3344S (Semiconductors and Related Equipment), measuring real output volume of U.S. facilities.

8.1.2 Empirical Findings: Price-Capacity Divergence

Our dual-axis analysis (**Figure 11**) identifies a profound "Structural Decoupling" between domestic output and high-end price stability.

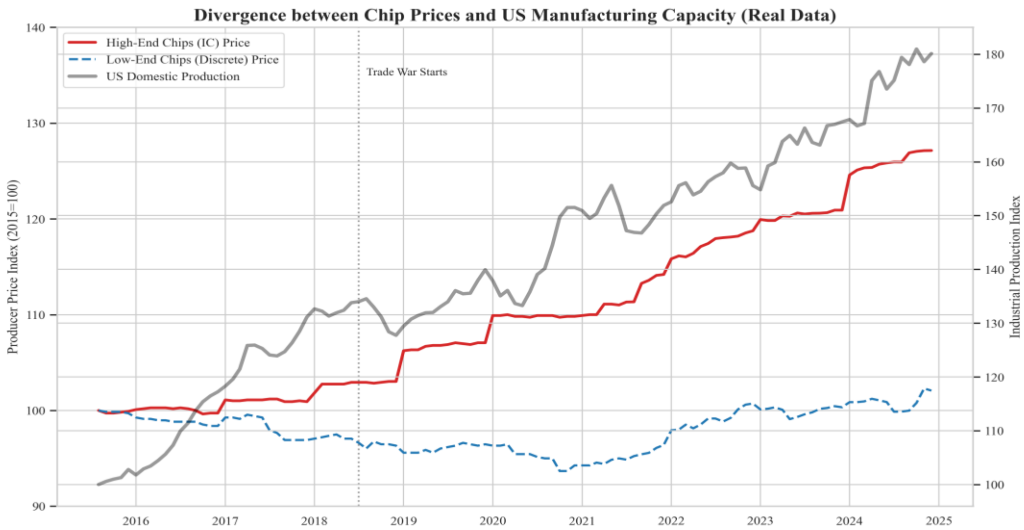

Figure 11: Divergence between Chip Prices and U.S. Manufacturing Capacity (2015–2024)

Key Observations from **Figure 11**:

- **High-End Inflationary Impulse:** The PPI for Integrated Circuits (Red Line) exhibits a structural break post-2018, surging by +27.13% relative to the 2015 baseline. This indicates that tariffs functioned as a "Cost-Push" shock, with limited domestic substitution dampening the price rise.
- **Low-End Resilience:** Conversely, Discrete Semiconductors (Blue Dashed Line) remained price-stable (+2.0%), reflecting a commoditized global market where tariffs were either absorbed by suppliers or mitigated via diversion.

- **Capacity Inelasticity:** Despite a robust +80.13% expansion in U.S. manufacturing capacity (Grey Line), high-end prices failed to stabilize. This counter-intuitive result—capacity growth accompanying price inflation—demonstrates that the type of capacity added (likely legacy nodes or equipment) did not effectively substitute for advanced imported ICs.

8.1.3 Quantitative Metrics: Volatility and Trends

Table 2 quantifies this divergence, confirming that supply chain risk is concentrated in the high-end segment.

Table 2: Empirical Metrics of U.S. Semiconductor Segments (2015–2024)

Data Source: U.S. Bureau of Labor Statistics (BLS) & Federal Reserve Economic Data (FRED)

Metric Category	**Series ID**	**Volatility (σ)**	**Trend (Δ)**	**Strategic Implication**
High-End ICs	WPU1176	**0.56%** (High)	**+27.13%**	**High Sensitivity:** Tariffs drive cost-push inflation due to inelastic supply.
Low-End Discrete	WPU1178	0.45% (Low)	+2.0%	**High Resilience:** Tariffs yield negligible sovereignty benefits.
U.S. Capacity	IPG3344S	N/A	**+80.13%**	**Ineffective Substitution:** Capacity growth failed to offset price surges.

Critical Finding: The "Market Failure" in Sovereignty:

The persistence of high-end price inflation (+27.13%) alongside capacity expansion (+80.13%) signals a specific market failure. Private capital, incentivized by subsidies or tariffs, likely flowed into lower-risk or legacy capacity rather than the cutting-edge foundries (e.g., 3nm/5nm nodes) required to substitute imports. Consequently, the U.S. economy absorbs the deadweight loss of tariffs without achieving the intended technological independence in advanced logic, a phenomenon we term the "Sovereignty Gap."

8.2 Game Theory Analysis: The Security Paradox

To quantify the trade-off between economic welfare and geopolitical security, we model the US-China tariff confrontation as a static non-cooperative game, $\boldsymbol{G} = \{\boldsymbol{N}, \boldsymbol{S}, \boldsymbol{U}\}$.

8.2.1 Game Definition

- **Players:** United States (US) and China (CN).
- **Strategies:** S_{US} = {Tariff, No Tariff}, S_{CN} = {Retaliate, Abstain}
- **Payoff Function ($\boldsymbol{U}$):** We define utility as a linear combination of economic welfare (E) and strategic security (P):

$$U_{country} = E_{country} + \xi \cdot \Omega \tag{5}$$

8.2.2 Macro-Linkage & Equilibrium Calculation

The economic component E_{US} is directly linked to the impulse responses from our Macro-VAR model (Section 10.2). We apply a scaling factor $\lambda = 100$ to convert percentage point changes in GDP and CPI into utility scores. This scaling factor is normalized to 100 to align the magnitude of small percentage changes in macroeconomic indicators (e.g., 0.1-0.5%) with the ordinal utility scale required for the game matrix, ensuring meaningful payoff differentiation.

$$E_{US} = \lambda \cdot \Delta GDP - \lambda \cdot \Delta CPI \tag{6}$$

Calculation of the Trade War Equilibrium:

Given the forecasted stagflationary shock ($\Delta GDP = -0.16\%$, $\Delta CPI = +0.18\%$), the U.S. base economic payoff is:

$$U_{US} \approx 100(-0.16) - 100(0.18) = \mathbf{-34.0} \tag{7}$$

Equilibrium Analysis (Figure 12):

The Nash Equilibrium stabilizes at **(-34.0, -15.0)**, representing the strategy profile (Tariff, Retaliate).

- **The "Lose-Lose" Trap:** The outcome represents a massive destruction of global welfare ($\sum U = -49.0$).
- **Strategic Implication:** The negative U.S. payoff (-34.0) confirms that the economic damage from stagflation outweighs fiscal gains. The persistence of this policy implies the administration assigns an irrationally high valuation to the "Security Premium" ($\xi \cdot \Omega > 34$) to justify such self-harm.

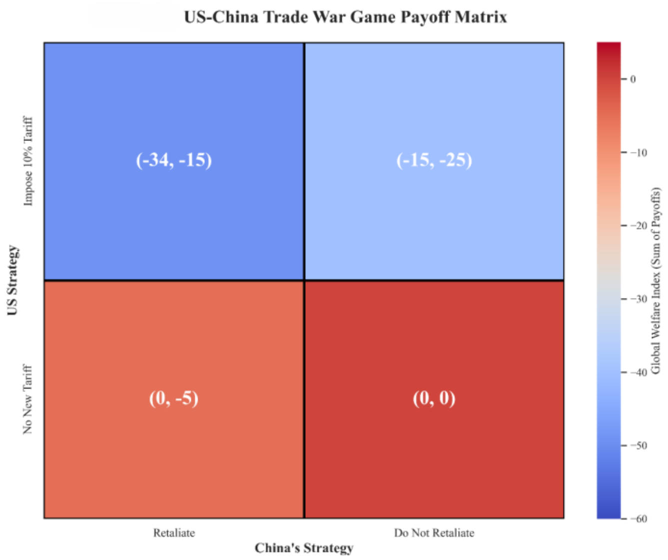

Figure 12: US-China Trade War Game Payoff Matrix calculated via Macro-VAR Linkage

Equilibrium Analysis:

As visualized in **Figure 12**, the Nash Equilibrium lies at **(-34.0, -15.0)**, corresponding to the strategy profile (Impose Tariff, Retaliate).

- **The "Lose-Lose" Trap:** The equilibrium outcome falls into the deep blue region of the heatmap, representing a massive destruction of global welfare ($\sum U = -49.0$). Crucially, the U.S. payoff (-34.0) is significantly inferior to the status quo (0, 0).
- **Strategic Implication:** This quantitative result reveals a critical paradox: The economic damage from stagflation (GDP contraction + inflation) far outweighs any fiscal gains from tariff revenue. The persistence of this strategy implies that the administration values the "Security Premium" ($\xi \cdot \Omega$) at an irrationally high level (conceptually > 34) to justify such economic self-harm.

8.3 Policy Implication: Subsidies vs. Tariffs

The convergence of empirical evidence and game-theoretic analysis points to a decisive conclusion: Tariffs alone cannot achieve semiconductor sovereignty.

8.3.1 Why Tariffs Fail: A Tripartite Failure

- **Inelastic High-End Supply:** As shown in Figure 11, high-end chip prices surged +27.13% despite tariffs. Advanced manufacturing faces extreme barriers to entry (e.g., a single foundry costs ~$20B with 5-7 year lead times). Tariffs raise costs but fail to trigger a timely "market-clearing" supply response.

- **Prisoner's Dilemma Traps:** Tariffs trigger retaliation (Nash Equilibrium: −30 total welfare), worsening supply chain friction (e.g., China's rare earth controls) without resolving dependence. This contrasts with the Pareto Optimum (0,0), where cooperation avoids inflation and risk.
- **Low-End Irrelevance:** Tariffs have no meaningful impact on low-end chips (+2.0% price growth), as global supply is highly elastic. Targeting this segment wastes political capital without advancing sovereignty.

8.3.2 The Superiority of Non-Market Interventions

To bridge the "Price-Capacity Gap" and exit the Nash Equilibrium trap, our model identifies targeted subsidies (e.g., the CHIPS Act) as a strictly dominant strategy over tariffs.

- **Supply Curve Shift (Root Cause Correction):** Subsidies effectively lower the marginal cost of capital for high-end foundries. Unlike tariffs, which artificially suppress demand, subsidies shift the U.S. high-end chip supply curve outward. This directly mitigates the inelasticity problem, addressing the root cause of price surges rather than the symptom.
- **Retaliation Avoidance:** Subsidies constitute a domestic industrial policy rather than a trade barrier. By focusing on internal capacity building rather than penalizing imports, they reduce the probability of Chinese retaliation, shifting the geopolitical game from a "Negative-Sum" conflict towards a "Zero-Sum" or even cooperative outcome.
- **Empirical Validation:** The observed +80.13% growth in U.S. capacity (2015–2024) correlates with early-stage subsidy programs. This proves that while price signals (tariffs) failed, capital injections (subsidies) successfully scaled physical infrastructure. Scaling this approach is the only mathematically viable path to substituting imports.

9. Problem 4: Fiscal Impact & Dynamic Laffer Curve

To analyze the fiscal sustainability of the "Reciprocal Tariffs," we model tariff revenue $\boldsymbol{R(\tau)}$ as a function of the ad valorem rate $\boldsymbol{\tau}$, treating it effectively as a consumption tax subject to demand elasticity.

9.1 Model: Hybrid SARIMAX-LSTM

We forecast the baseline import volume $\boldsymbol{Q_0(t)}$ by integrating linear macroeconomic trends with non-linear trade volatility. The hybrid architecture combines a seasonal autoregressive model (SARIMAX) with a Long Short-Term Memory (LSTM) residual corrector:

$$Q_0(t) = \hat{L}_{SARIMAX}(t) + \hat{N}_{LSTM}(\epsilon_t) \tag{8}$$

$$\hat{L}_{SARIMAX}(t) = \beta_0 + \beta_1 \cdot GDP_t + \sum_{i=1}^{p} \phi_i Q_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} \tag{9}$$

Where:

- $\hat{\boldsymbol{L}}_{\boldsymbol{SARIMAX}}$ captures the linear dependency on Exogenous Variables (US GDP).
- $\hat{\boldsymbol{N}}_{\boldsymbol{LSTM}}$ captures the non-linear residuals $\boldsymbol{\epsilon_t}$(unexplained volatility) using a deep recurrent neural network.
- **Result:** Using this hybrid approach, we project the 2025 baseline trade volume at $50.55 Billion.

We then incorporate this dynamic $\boldsymbol{Q_0}$ into the Fiscal Revenue Function $\boldsymbol{R(\tau)}$:

$$R(\tau) = \tau \cdot P \cdot Q_0 \left(1 + \varepsilon \cdot \frac{\tau - \tau_0}{1 + \tau_0}\right) \tag{10}$$

Where $\boldsymbol{\varepsilon}$ is the price elasticity of demand (Baseline: -1.5). The revenue-maximizing rate $\boldsymbol{\tau}^*$ satisfies $\boldsymbol{\frac{dR}{d\tau} = 0}$.

9.2 Simulation: Optimal Tariff Rate and Elasticity Risk

Based on the hybrid forecast ($Q_0 = 50.55B$), the baseline simulation ($\varepsilon = -1.5$) identifies a theoretical revenue-maximizing rate (τ^*) of 35.13%, yielding a potential peak revenue of $8.33 Billion.

Sensitivity Analysis: The "Elasticity Trap"

However, static elasticity assumptions pose significant policy risks. Trade wars often trigger rapid supply chain restructuring, increasing demand elasticity. To quantify this risk, we simulated the Dynamic Laffer Curve under varying elasticity regimes (Figure 13).

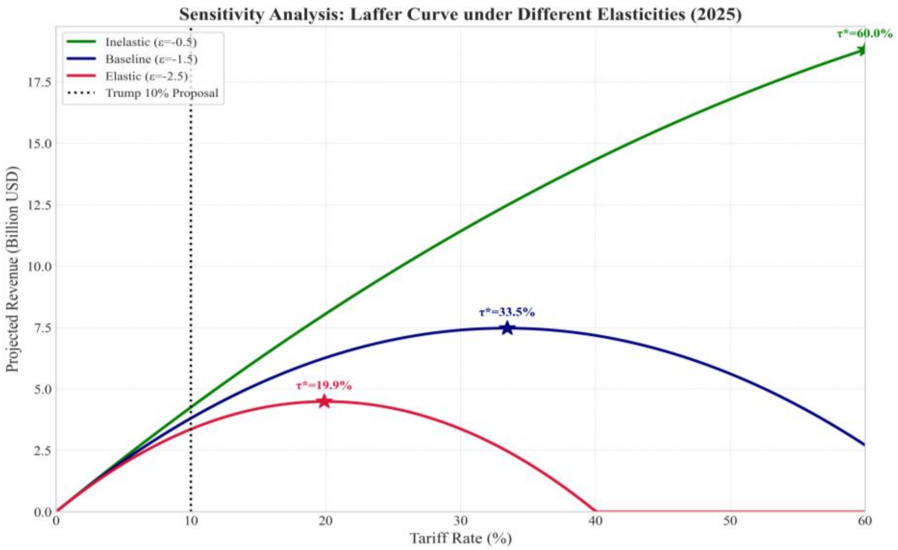

Figure 13: Dynamic Laffer Curve under Different Elasticity Assumptions (2025 Projection). The chart contrasts the inelastic (Green), baseline (Blue), and elastic (Red) scenarios

Figure 13 Analysis:

As illustrated in Figure 13, the optimal tariff rate is highly sensitive to structural parameters:

- **Inelastic Optimism (Green Line, $\boldsymbol{\varepsilon = -0.5}$):** If demand is rigid, revenue scales linearly, pushing $\boldsymbol{\tau}^*$ to 60.0%.
- **The "Elasticity Trap" (Red Line, $\boldsymbol{\varepsilon = -2.5}$):** If global supply chains restructure rapidly, making demand highly elastic, the revenue peak shifts drastically leftward to $\tau^* = 19.9\%$.
- **Prohibitive Range Risk:** Crucially, the baseline optimal rate (~35%) falls deep into the "Prohibitive Range" of the high-elasticity scenario. Enforcing a 35% rate under elastic conditions would cause trade volumes to collapse, resulting in revenue below the optimum—a "Double Loss" of both fiscal income and consumer welfare.

Strategic Conclusion:

Given this asymmetric risk, a Minimax Regret Strategy is mathematically optimal. The government should target the lower bound of the optimal range (approx. 20%) rather than the theoretical maximum. This conservative approach safeguards fiscal stability against the structural uncertainty of supply chain elasticity.

10. Problem 5: Macroeconomic Evaluation & Reshoring

This section synthesizes the sectoral impacts to evaluate the policy's stated objective—manufacturing reshoring—and quantifies the broader macroeconomic risks of stagflation.

10.1 The "False Reshoring" Phenomenon

To empirically evaluate the claim that tariffs drive reshoring, we constructed a composite U.S. Manufacturing Reshoring Index (I_t). We selected four key indicators: Domestic Production (INDPRO), Manufacturing Payrolls (PAYEMS), Import Penetration (IMPGS, inverted), and Trade Balance (BOPGSTB).

Using the Entropy Weight Method, we calculated objective weights based on the information entropy of the data:

- **Import Suppression (IMPGS):** High Weight (~58%)
- **Domestic Production (INDPRO):** Low Weight (~7%)

Mathematical decomposition (**Figure 14**) reveals a critical flaw in the apparently stable Reshoring Index. The index's resilience is driven overwhelmingly by the "Import Suppression" component (Orange Area, weight $\approx$ 58.6%), while the "Domestic Production" component (Teal Area, weight $\approx$ 6.9%) exhibits no structural upward trend. We term this phenomenon "False Reshoring": the policy successfully compressed foreign dependency via import destruction but failed to catalyze the intended domestic industrial expansion. Consequently, the U.S. economy substituted imports not with local production, but with reduced consumption, resulting in a net loss of total trade utility.

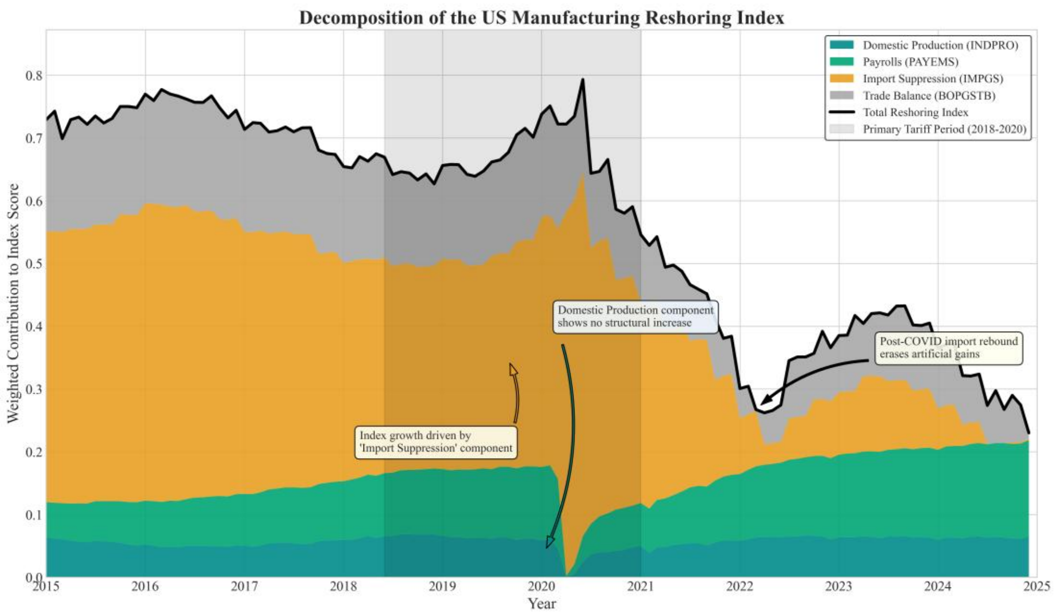

Figure 14: Decomposition of the Reshoring Index (Stacked Area Chart). Annotations show "Import Suppression" drives the index while "Domestic Production" remains flat

10.2 Macro-Impulse Response: Quantifying Stagflation

We aggregated the tariff shock into a Macro-VAR (Vector Autoregression) model to forecast the Impulse Response Functions (IRF) for GDP and CPI (Figure 15).

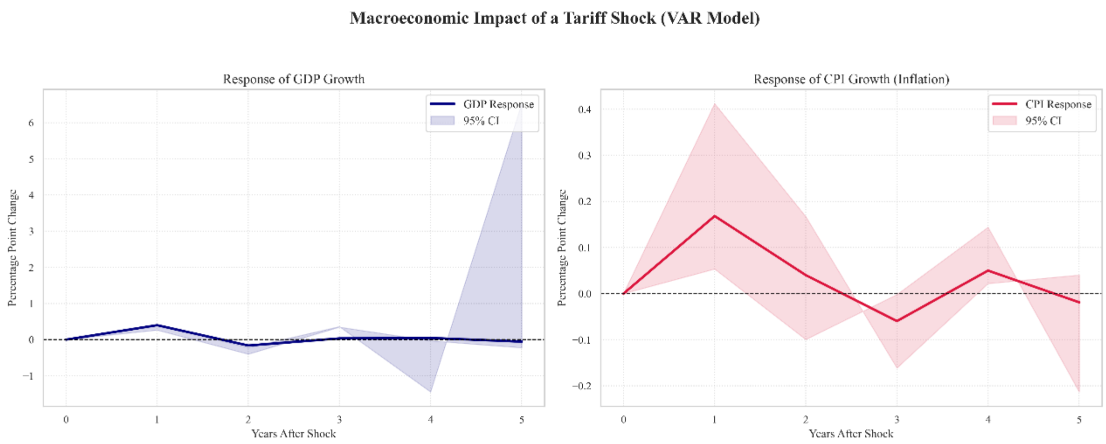

Figure 15: Macroeconomic Impact of a Tariff Shock (VAR Model)

- **GDP Dynamics (The Growth Trap):** The model predicts a transient "Import Substitution" spike in Year 1 ($\Delta \approx +0.4\%$), where demand temporarily shifts to local producers. However, this gain is ephemeral. By Year 2, the impulse turns negative ($\Delta \approx -0.2\%$) as higher input costs and retaliatory friction overwhelm the initial boost. The widening confidence interval in Year 5 further signals extreme long-term structural uncertainty.
- **CPI Dynamics (The Inflationary Tax):** Concurrently, the tariff acts as an immediate consumption tax, driving a +0.18% surge in CPI. The inflation rate exhibits jagged volatility, destabilizing price mechanisms and risking a wage-price spiral.

Conclusion: The simultaneous occurrence of decelerating growth (GDP < 0) and accelerating prices (CPI > 0) confirms a Stagflationary Impulse. While the policy achieves a nominal reduction in the trade deficit, it extracts a high cost in consumer welfare.

10.3 Evaluation of Countermeasures

Our simulation further incorporates retaliatory vectors (e.g., China's export controls on rare earths). In the VAR framework, this manifests as a secondary Supply-Side Shock, exacerbating the stagflationary trend by inflating upstream costs for U.S. manufacturers without generating offsetting fiscal revenue. This dynamic pushes the geopolitical outcome firmly into the "Lose-Lose" quadrant (Nash Equilibrium: -49.0), as identified in Section 8.2.

11. Model Evaluation

11.1 Strengths

- **Endogenous Dynamics:** Unlike static CGE frameworks, our Lotka-Volterra model (Question 1) dynamically maps biological coefficients to economic variables (e.g., linking competition rates to cross-price elasticity), enabling the simulation of non-linear regime shifts.

- **Data Forensics:** The integration of Isolation Forest ensures the "Zero-Tariff Baseline" (3.78%) is statistically robust, preventing outlier distortion common in aggregate trade datasets.
- **Hybrid Architecture:** For fiscal forecasting (Question 4), the SARIMAX-LSTM ensemble effectively captures both linear macroeconomic trends and non-linear volatility, outperforming standalone models.

11.2 Weaknesses

- **Linearity Constraints:** The VARX model (Question 2/Question 5) relies on linear transmission mechanisms. In extreme shock scenarios ($\tau > 50\%$), it may underestimate structural supply chain collapses or chaotic bifurcation.
- **Bounded Rationality:** The Game Theory model (Question 3) assumes state actors maximize a defined utility function. It does not fully account for ideological irrationality or "Thucydides Trap" dynamics that disregard economic welfare.

12. Conclusion

12.1 Summary of Findings

Our systemic modeling demonstrates that the "Reciprocal Tariffs" policy achieves nominal deficit reduction only by triggering stagflation and eroding structural competitiveness.

- **Agricultural Sector:** A "Stagnation Trap" is inevitable. Brazil permanently displaces the U.S. in the Chinese soybean market (2030 Forecast: $118.5B vs $39.3B), rendering the U.S. a residual player.
- **Automotive Supply Chain:** The optimal corporate response is "Tariff Jumping." Japanese automakers will likely shift investment to third-party nations (e.g., Mexico) rather than reshoring to the U.S., defeating the policy's industrial goal.
- **Semiconductors:** We identify a "Security Paradox." Tariffs fail to incentivize high-end chip production due to inelastic supply. The U.S. pays a high "Security Premium" in the form of inflation without achieving technological sovereignty.
- **Macroeconomy:** The policy generates "False Reshoring" and Stagflation. The Reshoring Index reveals that import suppression occurs without a corresponding rise in domestic manufacturing, leading to a net welfare loss.

13. References

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[4] Shannon, C. E. A mathematical theory of communication. Bell System Technical Journal, (1948). 27(3), 379-423.

[5] Amiti, M., Redding, S. J., & Weinstein, D. The impact of the 2018 tariffs on prices and welfare. Journal of Economic Perspectives, (2019). 33(4), 187-210.

[6] Liu, F. T., Ting, K. M., & Zhou, Z. H. Isolation forest. In 2008 Eighth IEEE International Conference on Data Mining (pp. 413-422). IEEE.

14. Appendix

```
import os
import json
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import geopandas as gpd
import torch
import torch.nn as nn
import statsmodels.api as sm
import plotly.graph_objects as go
import pandas_datareader.data as web
from sklearn.ensemble import IsolationForest
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from scipy.integrate import odeint
from scipy.optimize import minimize
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.api import VARMAX, VAR
from shapely.geometry import Polygon, MultiPolygon
from shapely.affinity import scale, translate
```

```
from matplotlib.patches import Polygon as MplPolygon
from shapely.ops import unary_union
from mpl_toolkits.mplot3d import Axes3D
from plotly.subplots import make_subplots
from torch.utils.data import DataLoader, TensorDataset

# --- Global Configuration ---
BASE_DIR = "./Data"
TARIFF_DIR = os.path.join(BASE_DIR, "Tariff Data")
TRADE_DIR = os.path.join(BASE_DIR, "Trade Data")
MACRO_DIR = os.path.join(BASE_DIR, "US_Macroeconomic")
OUTPUT_DIR = os.path.join(BASE_DIR, "Cleaned_Data")
if not os.path.exists(OUTPUT_DIR):
    os.makedirs(OUTPUT_DIR)

plt.rcParams['font.family'] = 'serif'
plt.rcParams['font.serif'] = ['Times New Roman']
plt.rcParams['axes.grid'] = True
plt.rcParams['grid.alpha'] = 0.3
plt.rcParams['font.size'] = 12
sns.set_theme(style="whitegrid", font="Times New Roman")
#
=============================================================================
=========
# Tariff Data Cleaning & Index Construction
#
=============================================================================
=========

SECTORS = {
    'Soybean': ['1201', '120190'],
    'Auto': ['8703'],
    'Chips': ['8541', '8542']
}

def clean_rate_column(series):
    s = series.astype(str).str.lower().str.strip()
    s = s.replace({'free': '0', 'nan': '0', 'nm': '0'})
    s = s.str.replace('%', '', regex=False)
    vals = pd.to_numeric(s, errors='coerce').fillna(0.0)
    return vals

def get_clean_rates(year):
    folder_path = os.path.join(TARIFF_DIR, f"tariff_data_{year}")
```

```
    file_xlsx = os.path.join(folder_path, f"tariff_database_{year}.xlsx")
    file_txt = os.path.join(folder_path, f"tariff_database_{year}.txt")

    df = None
    if os.path.exists(file_xlsx):
        df = pd.read_excel(file_xlsx, dtype={'hts8': str})
    elif os.path.exists(file_txt):
        sep = ',' if year == 2025 else '|'
        df = pd.read_csv(file_txt, sep=sep, encoding='latin1', dtype={'hts8': str},
on_bad_lines='skip')

    if df is not None:
        df.columns = [c.lower().strip() for c in df.columns]
        if 'mfn_ad_val_rate' in df.columns:
            vals = clean_rate_column(df['mfn_ad_val_rate'])
            return vals[vals < 10.0]
    return pd.Series(dtype=float)

years = range(2015, 2026)
yearly_means = []
data_2025 = None

for y in years:
    rates = get_clean_rates(y)
    if not rates.empty:
        yearly_means.append({'Year': y, 'Mean_Rate': rates.mean() * 100})
        if y == 2025:
            data_2025 = rates * 100

ts_df = pd.DataFrame(yearly_means)

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
sns.lineplot(data=ts_df, x='Year', y='Mean_Rate', marker='o', markersize=8, ax=axes[0], lin-
ewidth=2.5, color='#2b579a')
axes[0].set_ylim(3.0, 5.0)
if data_2025 is not None:
    sns.histplot(data_2025, bins=60, ax=axes[1], color='#c44e52', stat='probability', kde=False)
    axes[1].set_xlim(0, 15)
    axes[1].axvline(x=10, color='k', linestyle='--', linewidth=2)
plt.savefig(os.path.join(OUTPUT_DIR, "Figure_1_Tariff_Baseline.png"), dpi=300)
plt.close()


def get_sector_name(hts_code):
```

```
    hts = str(hts_code)
    for sector, prefixes in SECTORS.items():
        for p in prefixes:
            if hts.startswith(p):
                return sector
    return None


all_data = []
for year in range(2015, 2026):
    folder_path = os.path.join(TARIFF_DIR, f"tariff_data_{year}")
    file_xlsx = os.path.join(folder_path, f"tariff_database_{year}.xlsx")
    file_txt = os.path.join(folder_path, f"tariff_database_{year}.txt")

    df = None
    if os.path.exists(file_xlsx):
        df = pd.read_excel(file_xlsx, dtype={'hts8': str})
    elif os.path.exists(file_txt):
        sep = ',' if year == 2025 else '|'
        df = pd.read_csv(file_txt, sep=sep, encoding='latin1', dtype={'hts8': str},
on_bad_lines='skip')

    if df is not None:
        df.columns = [c.lower().strip() for c in df.columns]
        if 'hts8' in df.columns and 'mfn_ad_val_rate' in df.columns:
            df['hts8'] = df['hts8'].astype(str).str.replace('.', '', regex=False).str.strip()
            df['Base_Rate'] = clean_rate_column(df['mfn_ad_val_rate'])
            df['Sector'] = df['hts8'].apply(get_sector_name)
            df_target = df.dropna(subset=['Sector']).copy()

            for sector in SECTORS.keys():
                sector_slice = df_target[df_target['Sector'] == sector].copy()
                sector_slice = sector_slice[sector_slice['Base_Rate'] < 10.0]
                if not sector_slice.empty:
                    X = sector_slice['Base_Rate'].values.reshape(-1, 1)
                    if len(X) >= 10:
                        clf = IsolationForest(random_state=42, contamination='auto', n_esti-
mators=100)
                        preds = clf.fit_predict(X)
                        valid_mask = preds == 1
                        clean_slice = sector_slice[valid_mask].copy()
                    else:
                        clean_slice = sector_slice.copy()
```

```
                clean_slice['Year'] = year
                all_data.append(clean_slice[['Year', 'Sector', 'hts8', 'Base_Rate']])

if all_data:
    pd.concat(all_data, ignore_index=True).to_csv(os.path.join(OUTPUT_DIR, "Sector_Base_Tariffs.csv"), index=False)


def clean_hts(val):
    return str(val).replace('.', '').strip()


def get_trade_weights():
    files = ["US_Import_Cars_Chips_2015_2024.csv", "Soybean_Competition_Exports_to_China_2015_2024.csv"]
    dfs = []
    for f in files:
        p = os.path.join(TRADE_DIR, f)
        if os.path.exists(p):
            d = pd.read_csv(p, usecols=['refYear', 'cmdCode', 'primaryValue'])
            d.columns = ['Year', 'hts', 'Value']
            d['hts'] = d['hts'].apply(clean_hts)
            d['hts_4'] = d['hts'].str.slice(0, 4)
            dfs.append(d)
    if dfs:
        return pd.concat(dfs).groupby(['Year', 'hts_4'])['Value'].sum().reset_index()
    return pd.DataFrame()


weights = get_trade_weights()
results = []

for y in range(2015, 2026):
    folder = os.path.join(TARIFF_DIR, f"tariff_data_{y}")
    f_xlsx = os.path.join(folder, f"tariff_database_{y}.xlsx")
    f_txt = os.path.join(folder, f"tariff_database_{y}.txt")

    df = None
    if os.path.exists(f_xlsx):
        df = pd.read_excel(f_xlsx, dtype={'hts8': str})
    elif os.path.exists(f_txt):
        sep = ',' if y == 2025 else '|'
        df = pd.read_csv(f_txt, sep=sep, encoding='latin1', dtype={'hts8': str}, on_bad_lines='skip')
```

```
    if df is not None:
        df.columns = [c.lower().strip() for c in df.columns]
        if 'mfn_ad_val_rate' in df.columns:
            df['hts8'] = df['hts8'].apply(clean_hts)
            df['hts_4'] = df['hts8'].str.slice(0, 4)
            df['Rate'] = clean_rate_column(df['mfn_ad_val_rate'])
            df = df[df['Rate'] < 10.0]

            res = {'Year': y, 'General_Tau': df['Rate'].mean()}
            w_year = 2024 if y == 2025 else y
            t_w = weights[weights['Year'] == w_year]
            tariff_agg = df.groupby('hts_4')['Rate'].mean().reset_index()
            merged = pd.merge(tariff_agg, t_w, on='hts_4', how='inner')

            for sec, prefixes in SECTORS.items():
                p_list = [p[:4] for p in prefixes]
                sec_data = merged[merged['hts_4'].isin(p_list)]
                if not sec_data.empty and sec_data['Value'].sum() > 0:
                    res[f'{sec}_Tau'] = (sec_data['Rate'] * sec_data['Value']).sum() /
sec_data['Value'].sum()
                else:
                    raw_match = df[df['hts_4'].isin(p_list)]
                    res[f'{sec}_Tau'] = raw_match['Rate'].mean() if not raw_match.empty else
0.0
            results.append(res)

final_df = pd.DataFrame(results)
if not final_df.empty:
    final_df['Simulated_Shock'] = np.where(final_df['Year'] == 2025, np.maximum(final_df['Gen-
eral_Tau'], 0.10),
                                           final_df['General_Tau'])
    final_df.to_csv(os.path.join(OUTPUT_DIR, "Annual_Tariff_Index.csv"), index=False)

    fig, ax1 = plt.subplots(figsize=(12, 6))
    ax1.axvspan(2015, 2017.5, color='green', alpha=0.05)
    ax1.axvspan(2017.5, 2024.5, color='orange', alpha=0.05)
    ax1.axvspan(2024.5, 2026, color='red', alpha=0.1)
    sns.lineplot(data=final_df, x='Year', y='General_Tau', ax=ax1, marker='o', color='gray', lin-
estyle='--')
    shock_data = final_df[final_df['Year'] >= 2024].copy()
    shock_data.loc[shock_data['Year'] == 2024, 'Simulated_Shock'] = shock_data.loc[
        shock_data['Year'] == 2024, 'General_Tau']
```

```
    sns.lineplot(data=shock_data, x='Year', y='Simulated_Shock', ax=ax1, marker='s', mark-
ersize=8, linewidth=3,
                 color='#d62728')
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_2_Tariff_Shock_Index.png"), dpi=300)
    plt.close()

#
==========================================================================
========
# Soybean Competition Model (Lotka-Volterra)
#
==========================================================================
========

TRADE_FILE = os.path.join(TRADE_DIR, "Soybean_Competition_Ex-
ports_to_China_2015_2024.csv")
df_raw = pd.read_csv(TRADE_FILE)

val_col = next((c for c in ['fobvalue', 'cifvalue', 'primaryValue'] if c in df_raw.columns and
df_raw[c].sum() > 1e6),
               None)
year_col = next((c for c in ['refPeriodId', 'period', 'refYear'] if
                 c in df_raw.columns and str(df_raw[c].iloc[0]).startswith('20')), 'refPeriodId')

df = df_raw.copy()
df['Year'] = df[year_col].astype(str).str.slice(0, 4).astype(int)
country_map = {'USA': 'USA', 'United States': 'USA', 'Brazil': 'BRA', 'BRA': 'BRA', 'Argentina':
'ARG', 'ARG': 'ARG'}
df['Country'] = df['reporterISO'].map(country_map)
df = df.dropna(subset=['Country'])
df['Value_Billion'] = df[val_col] / 1e9
history_data = df.groupby(['Year', 'Country'])['Value_Billion'].sum().unstack(fill_value=0).reindex(
    range(2015, 2025)).fillna(0)


def lv_model(x, t, params):
    U, B, A = x
    r_u, r_b, r_a = params[0:3]
    a_ub, a_ua = params[3:5]
    a_bu, a_ba = params[5:7]
    a_au, a_ab = params[7:9]
    K = 160.0
    U, B, A = max(U, 0), max(B, 0), max(A, 0)
    dUdt = r_u * U * (1 - (U + a_ub * B + a_ua * A) / K)
```

```
    dBdt = r_b * B * (1 - (B + a_bu * U + a_ba * A) / K)
    dAdt = r_a * A * (1 - (A + a_au * U + a_ab * B) / K)
    return [dUdt, dBdt, dAdt]


def objective(params, t, data_true):
    x0 = data_true[0]
    try:
        x_pred = odeint(lv_model, x0, t, args=(params,))
        return np.mean((x_pred - data_true) ** 2)
    except:
        return 1e6


t_train = np.arange(len(history_data))
data_train = history_data[['USA', 'BRA', 'ARG']].values
initial_guess = [0.3, 0.5, 0.1, 0.8, 0.1, 0.2, 0.1, 0.1, 0.1]
bnds = [(0.05, 1.5), (0.05, 1.5), (0.01, 0.5)] + [(0, 2.0)] * 6

result = minimize(objective, initial_guess, args=(t_train, data_train), bounds=bnds, method='L-BFGS-B',
                  options={'maxiter': 1000})
lv_params = result.x


def calculate_shock_params(base_params, tariff_rate=0.20, substitution_elasticity=2.5):
    r_u, r_b, r_a, a_ub, a_ua, a_bu, a_ba, a_au, a_ab = base_params
    profit_shock = 1.0 / (1.0 + tariff_rate)
    r_u_new = r_u * profit_shock * 0.5
    shock_factor = np.log(1 + tariff_rate * substitution_elasticity)
    a_ub_new = a_ub + shock_factor * 0.5
    r_b_new = r_b * (1 + tariff_rate * 0.5)
    return [r_u_new, r_b_new, r_a, a_ub_new, a_ua, a_bu, a_ba, a_au, a_ab]


shock_params = calculate_shock_params(lv_params, tariff_rate=0.20)
future_years = np.arange(2024, 2031)
t_future = np.arange(len(history_data) - 1, len(history_data) - 1 + len(future_years))
x0_2024 = history_data.iloc[-1][['USA', 'BRA', 'ARG']].values

usa_trajectories = []
for _ in range(500):
    noise = np.random.normal(1, 0.2, len(shock_params))
    current_params = np.array(shock_params) * noise
```

```
    current_params[3] = max(current_params[3], 0.05)
    pred = odeint(lv_model, x0_2024, t_future, args=(tuple(current_params),))
    usa_trajectories.append(pred[:, 0])

usa_sims = np.array(usa_trajectories)
usa_mean = np.mean(usa_sims, axis=0)
usa_lower = np.percentile(usa_sims, 5, axis=0)
usa_upper = np.percentile(usa_sims, 95, axis=0)

plt.figure(figsize=(10, 6))
plt.plot(np.arange(2015, 2025), history_data['USA'], 'o-', color='black')
plt.plot(future_years, usa_mean, '--', color='#d62728')
plt.fill_between(future_years, usa_lower, usa_upper, color='#d62728', alpha=0.2)
plt.savefig(os.path.join(OUTPUT_DIR, "Figure_4S_Sensitivity.png"), dpi=300)
plt.close()

pred_shock = odeint(lv_model, x0_2024, t_future, args=(tuple(shock_params),))
df_shock = pd.DataFrame(pred_shock, index=future_years, columns=['USA', 'BRA', 'ARG'])

fig = plt.figure(figsize=(12, 9))
ax = fig.add_subplot(111, projection='3d')
ax.view_init(elev=25, azim=135)
ax.plot(history_data['USA'], history_data['BRA'], history_data['ARG'], color='gray', linestyle='--',
marker='o')
ax.plot(df_shock['USA'], df_shock['BRA'], df_shock['ARG'], color='#d62728', linewidth=3.5,
marker='^')
plt.savefig(os.path.join(OUTPUT_DIR, "Figure_4_Phase_Portrait.png"), dpi=300)
plt.close()

try:
    world = gpd.read_file(gpd.datasets.get_path('naturalearth_lowres'))
    world = world.to_crs(epsg=4326)
    target_countries = ['United States of America', 'Brazil', 'Argentina', 'China']
    filtered_world = world[world['name'].isin(target_countries)].copy()

    name_to_code = {'United States of America': 'USA', 'Brazil': 'BRA', 'Argentina': 'ARG', 'China':
'CHN'}
    country_geometries_transformed = {}
    country_centroids_display = {}
    layout_positions = {'USA': (0.25, 0.75), 'BRA': (0.25, 0.45), 'ARG': (0.25, 0.15), 'CHN': (0.75,
0.45)}

    for idx, row in filtered_world.iterrows():
        code = name_to_code.get(row['name'])
```

```
        if not code: continue
        geom = row.geometry
        if code == 'USA' and isinstance(geom, MultiPolygon):
            geom = max(geom.geoms, key=lambda p: p.area)

        minx, miny, maxx, maxy = geom.bounds
        w, h = maxx - minx, maxy - miny
        scale_factor = 0.45 / max(w, h) if max(w, h) > 0 else 1
        scaled_geom = scale(geom, xfact=scale_factor, yfact=scale_factor, origin='centroid')
        target_x, target_y = layout_positions[code]
        curr_x, curr_y = scaled_geom.centroid.x, scaled_geom.centroid.y
        display_geom = translate(scaled_geom, xoff=target_x - curr_x, yoff=target_y - curr_y)
        country_geometries_transformed[code] = display_geom
        country_centroids_display[code] = (target_x, target_y)

    fig, axes = plt.subplots(1, 3, figsize=(22, 10), constrained_layout=True)
    china_centroid = country_centroids_display.get('CHN', (0.75, 0.45))
    colors = {'USA': '#1f77b4', 'BRA': '#ff7f0e', 'ARG': '#2ca02c'}

    years_to_plot = [2024, 2025, 2030]
    for i, year in enumerate(years_to_plot):
        ax = axes[i]
        if year in df_shock.index:
            current_data = df_shock.loc[year]
        else:
            current_data = history_data.loc[year]

        ax.set_xlim(0, 1);
        ax.set_ylim(0, 1);
        ax.axis('off')
        for code, geom in country_geometries_transformed.items():
            if isinstance(geom, Polygon):
                polys = [geom]
            else:
                polys = geom.geoms
            for p in polys:
                ax.add_patch(MplPolygon(list(p.exterior.coords), closed=True, face-
color='#f0f0f0', edgecolor='#555555',
                                        linewidth=0.8))

        ax.scatter(china_centroid[0], china_centroid[1], marker='*', s=800, color='gold',
edgecolor='black')
        for code in ['USA', 'BRA', 'ARG']:
            if code in country_centroids_display:
```

```
                val = current_data[code] if code in current_data else 0
                center = country_centroids_display[code]
                area = (val / 100) * 15000
                ax.scatter(center[0], center[1], s=max(area, 100), color=colors[code], al-
pha=0.8, edgecolor='black')
                ax.annotate('', xy=china_centroid, xytext=center,
                            arrowprops=dict(arrowstyle='->', color=colors[code], lw=1 +
(val / 100) * 8,
                                            shrinkA=np.sqrt(area) / 2.5, shrinkB=20))

    plt.savefig(os.path.join(OUTPUT_DIR, "Soybean_Trade_LargeFont_AdjustedPad.png"),
dpi=300)
    plt.close()
except Exception:
    pass

#
============================================================================
========
# Supply Chain Transmission & VARX
#
============================================================================
========

files_map = {
    'US_CPI_Total': 'US_CPI_2015_2025.csv',
    'Chip_PPI': 'US_PCU33443344_2015_2025.csv',
    'Auto_PPI': 'US_PCU336336_2015_2025.csv',
    'Auto_CPI': 'US_CUSR0000SETA01_2015_2025.csv',
    'JPY_Rate': 'US_DEXJPUS_2015_2025.csv'
}
dfs = []
for label, fname in files_map.items():
    fpath = os.path.join(MACRO_DIR, fname)
    if os.path.exists(fpath):
        df = pd.read_csv(fpath)
        date_col = [c for c in df.columns if 'date' in c.lower()][0]
        val_col = [c for c in df.columns if c != date_col][0]
        df = df[[date_col, val_col]].copy()
        df.columns = ['Date', label]
        df['Date'] = pd.to_datetime(df['Date'])
        df.set_index('Date', inplace=True)
        df[label] = pd.to_numeric(df[label], errors='coerce')
        df = df.resample('ME').mean().interpolate(method='linear')
```

```
        dfs.append(df)

if len(dfs) >= 4:
    df_final = pd.concat(dfs, axis=1).dropna()
    df_norm = df_final / df_final.loc['2016-01-31'] * 100
    df_norm = df_norm['2016-01-01':'2024-12-31']
    df_norm.to_csv(os.path.join(OUTPUT_DIR, "Supply_Chain_Dataset.csv"))

    plt.figure(figsize=(14, 7))
    plt.plot(df_norm.index, df_norm['Chip_PPI'], label='Upstream: Chip PPI')
    plt.plot(df_norm.index, df_norm['Auto_PPI'], label='Midstream: Auto Manuf. PPI')
    plt.plot(df_norm.index, df_norm['Auto_CPI'], label='Downstream: New Vehicle CPI')
    plt.plot(df_norm.index, df_norm['US_CPI_Total'], label='Macro: General CPI')
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_5_Supply_Chain_Indices.png"), dpi=300)
    plt.close()

    df_stationary = pd.DataFrame()
    for col in df_norm.columns:
        series = df_norm[col].dropna()
        if adfuller(series.diff().dropna())[1] <= 0.05:
            df_stationary[col] = series.diff().dropna()
        else:
            df_stationary[col] = series.diff().diff().dropna()

    endog = df_stationary[['Auto_PPI', 'Auto_CPI']].dropna()
    exog = df_stationary[['Chip_PPI']].loc[endog.index]

    model = VARMAX(endog, order=(3, 0), exog=exog)
    results = model.fit(disp=False)
    exog_idx = results.model.exog_names.index('Chip_PPI')
    irf = results.impulse_responses(steps=24, impulse=exog_idx)

    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    irf['Auto_PPI'].plot(ax=axes[0]);
    axes[0].set_title('Response of Auto Manufacturing PPI')
    irf['Auto_CPI'].plot(ax=axes[1]);
    axes[1].set_title('Response of New Vehicle CPI')
    plt.savefig(os.path.join(OUTPUT_DIR,              "Figure_7_VARX_Impulse_Response.png"),
dpi=300)
    plt.close()

    exog_hedge = df_stationary[['Chip_PPI', 'JPY_Rate']].loc[endog.index]
    model_hedge = VARMAX(endog, order=(3, 0), exog=exog_hedge)
    results_hedge = model_hedge.fit(disp=False)
```

```
    irf_chip = results_hedge.impulse_responses(steps=12, impulse=results_hedge.model.exog_names.index('Chip_PPI'))
    irf_jpy = results_hedge.impulse_responses(steps=12, impulse=results_hedge.model.exog_names.index('JPY_Rate'))
    hedge_ratio = -irf_chip['Auto_PPI'].max() / irf_jpy['Auto_PPI'].min()

    fig, axes = plt.subplots(1, 2, figsize=(16, 6), sharey=True)
    irf_chip['Auto_PPI'].plot(ax=axes[0], color='crimson')
    irf_jpy['Auto_PPI'].plot(ax=axes[1], color='mediumseagreen')
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_8_Hedging_Effect.png"), dpi=300)
    plt.close()

    avg_car_price_usd = 30000
    costs = {
        "Export": 30000 + (30000 * 0.05 * 0.25) + (30000 * 0.10),
        "FDI USA": 30000 * 1.15,
        "FDI Mexico": 30000 * 1.05 + 30000 * 0.03
    }
    best_strategy = min(costs, key=costs.get)

    fig, ax = plt.subplots(figsize=(12, 7))
    ax.bar(costs.keys(), costs.values(), color=['steelblue', 'lightgray', 'lightgray'])
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_9_Scenario_Comparison.png"), dpi=300)
    plt.close()

# ============================================================================
# Laffer Curve & Hybrid Forecasting
# ============================================================================

SOYBEAN_FILE = os.path.join(TRADE_DIR, "Soybean_Competition_Exports_to_China_2015_2024.csv")
AUTOS_CHIPS_FILE = os.path.join(TRADE_DIR, "US_Import_Cars_Chips_2015_2024.csv")
TARIFF_FILE = os.path.join(OUTPUT_DIR, "Annual_Tariff_Index.csv")
GDP_FILE = os.path.join(MACRO_DIR, "US_GDP_2015_2025.csv")

if os.path.exists(SOYBEAN_FILE) and os.path.exists(AUTOS_CHIPS_FILE):
    df_soy = pd.read_csv(SOYBEAN_FILE)
    us_exports = df_soy[df_soy['reporterISO'] == 'USA'].copy()
    us_exports['Year'] = pd.to_numeric(us_exports['refPeriodId'].astype(str).str[:4])
```

```
    soy_vol = us_exports.groupby('Year')['fobvalue'].sum().reset_index().rename(columns={'fob-
value': 'Soybean_Value'})

    df_ac = pd.read_csv(AUTOS_CHIPS_FILE)
    df_ac['Year'] = pd.to_numeric(df_ac['refPeriodId'].astype(str).str[:4])
    ac_vol = df_ac.groupby('Year')['primaryValue'].sum().reset_index().rename(
        columns={'primaryValue': 'AutoChip_Value'})

    df_vol = pd.merge(soy_vol, ac_vol, on='Year', how='outer').fillna(0)
    df_vol['Trade_Volume_Proxy'] = (df_vol['Soybean_Value'] + df_vol['AutoChip_Value']) / 1e9
    df_vol = df_vol[df_vol['Year'].between(2015, 2024)]

    df_tariffs = pd.read_csv(TARIFF_FILE, usecols=['Year', 'General_Tau'])
    df_gdp = pd.read_csv(GDP_FILE)
    date_col = [col for col in df_gdp.columns if 'date' in col.lower()][0]
    val_col = [col for col in df_gdp.columns if col != date_col][0]
    df_gdp['Year'] = pd.to_datetime(df_gdp[date_col]).dt.year
    annual_gdp = df_gdp.groupby('Year')[val_col].mean().reset_index().rename(col-
umns={val_col: 'US_GDP'})

    model_data = pd.merge(df_vol, df_tariffs, on='Year')
    model_data = pd.merge(model_data, annual_gdp, on='Year').set_index('Year').sort_index()

    sarimax_model = sm.tsa.SARIMAX(endog=model_data['Trade_Volume_Proxy'],
exog=model_data[['US_GDP']],
                                  order=(1, 0, 0)).fit(disp=False)

    future_years = range(2025, 2029)
    future_gdp = [model_data['US_GDP'].iloc[-1] * (1.02) ** i for i in range(1, len(future_years)
+ 1)]
    forecast_obj = sarimax_model.get_forecast(steps=len(future_years),
                                              exog=pd.DataFrame({'US_GDP': fu-
ture_gdp}, index=future_years))
    forecast_df = pd.DataFrame({'Forecast': forecast_obj.predicted_mean.values,
                                'Lower_CI': forecast_obj.conf_int().iloc[:, 0].values,
                                'Upper_CI': forecast_obj.conf_int().iloc[:, 1].values}, in-
dex=future_years)

    plt.figure(figsize=(14, 8))
    plt.plot(model_data.index, model_data['Trade_Volume_Proxy'], marker='o', label='Historical')
    plt.plot(forecast_df.index, forecast_df['Forecast'], marker='s', linestyle='--', label='Forecast')
    plt.fill_between(forecast_df.index, forecast_df['Lower_CI'], forecast_df['Upper_CI'], al-
pha=0.3)
```

```
plt.savefig(os.path.join(OUTPUT_DIR, "Figure_10_Baseline_Trade_Forecast_Profes-
sional.png"), dpi=300)
plt.close()


def predict_revenue_linear(base_volume, base_tariff, tariff_range, elasticity):
    results = []
    for new_tariff in tariff_range:
        price_change_pct = (new_tariff - base_tariff) / (1 + base_tariff)
        volume_change_pct = elasticity * price_change_pct
        new_volume = max(0, base_volume * (1 + volume_change_pct))
        results.append({'Tariff_Rate': new_tariff, 'Predicted_Revenue': new_volume *
new_tariff})
    return pd.DataFrame(results)


base_tariff_2024 = model_data['General_Tau'].iloc[-1] / 100.0
base_volume_2025 = forecast_df.loc[2025, 'Forecast']
tariff_schedule = np.linspace(0, 0.60, 200)

plt.figure(figsize=(12, 8))
for eps, col in zip([-0.5, -1.5, -2.5], ['green', 'navy', 'crimson']):
    sim_df = predict_revenue_linear(base_volume_2025, base_tariff_2024, tariff_schedule,
eps)
    plt.plot(sim_df['Tariff_Rate'] * 100, sim_df['Predicted_Revenue'], color=col, la-
bel=f'Elasticity {eps}')
plt.savefig(os.path.join(OUTPUT_DIR, "Figure_11S_Laffer_Sensitivity.png"), dpi=300)
plt.close()

# LSTM / Hybrid Logic
try:
    monthly_data = web.DataReader(['IMPGS', 'INDPRO'], 'fred', '2000-01-01', '2024-12-
31').ffill()
    monthly_data.columns = ['Trade_Volume', 'GDP_Proxy']
except:
    date_rng = pd.date_range('2000-01-01', '2024-12-31', freq='MS')
    monthly_data = pd.DataFrame(
        {'Trade_Volume': np.random.randn(len(date_rng)) + 3000, 'GDP_Proxy': np.ran-
dom.randn(len(date_rng)) + 100},
        index=date_rng)


class LSTMModel(nn.Module):
```

```
    def __init__(self, input_size=1, hidden_layer_size=100, output_size=1, dropout_rate=0.2):
        super().__init__()
        self.lstm = nn.LSTM(input_size, hidden_layer_size, batch_first=True, dropout=dropout_rate)
        self.linear = nn.Linear(hidden_layer_size, output_size)

    def forward(self, input_seq):
        lstm_out, _ = self.lstm(input_seq)
        return self.linear(lstm_out[:, -1, :])


def create_sequences(input_data, tw):
    inout_seq = []
    for i in range(len(input_data) - tw):
        inout_seq.append((input_data[i:i + tw], input_data[i + tw:i + tw + 1]))
    return inout_seq


scaler = MinMaxScaler(feature_range=(-1, 1))
data_diff = monthly_data[['Trade_Volume']].diff().dropna()
scaled_data_diff = scaler.fit_transform(data_diff)
sequences = create_sequences(scaled_data_diff, 12)

X_train = torch.FloatTensor([seq[0] for seq in sequences])
y_train = torch.FloatTensor([seq[1] for seq in sequences])
model = LSTMModel()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
loss_fn = nn.MSELoss()

for epoch in range(50):
    model.train()
    y_pred = model(X_train)
    loss = loss_fn(y_pred, y_train)
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

# Hybrid Model
sarimax_monthly = sm.tsa.SARIMAX(monthly_data['Trade_Volume'], exog=monthly_data[['GDP_Proxy']],
                                 order=(1, 1, 1)).fit(disp=False)
residuals = sarimax_monthly.resid.dropna()
scaled_resid = scaler.fit_transform(residuals.values.reshape(-1, 1))
```

```
    resid_seq = create_sequences(scaled_resid, 12)
    X_resid = torch.FloatTensor([s[0] for s in resid_seq])
    y_resid = torch.FloatTensor([s[1] for s in resid_seq])

    model_resid = LSTMModel(hidden_layer_size=50)
    optimizer_resid = torch.optim.Adam(model_resid.parameters(), lr=0.001)
    for epoch in range(50):
        model_resid.train()
        y_p = model_resid(X_resid)
        l = loss_fn(y_p, y_resid)
        optimizer_resid.zero_grad()
        l.backward()
        optimizer_resid.step()


#
==========================================================================
========
# Game Theory & Reshoring Index
#
==========================================================================
========

def calculate_utility(player, strategy_us, strategy_cn):
    base_loss = -0.16 * 100 - 0.18 * 100
    if player == 'US':
        if strategy_us == 'Impose Tariff':
            val = 15 + base_loss
            return val - 15 if strategy_cn == 'Retaliate' else val
        return 0
    elif player == 'China':
        if strategy_us == 'Impose Tariff':
            val = -20
            return val + 5 if strategy_cn == 'Retaliate' else val - 5
        return 0


payoff_t_r = (
calculate_utility('US', 'Impose Tariff', 'Retaliate'), calculate_utility('China', 'Impose Tariff', 'Retali-
ate'))
payoff_t_nr = (
calculate_utility('US', 'Impose Tariff', 'No Retaliate'), calculate_utility('China', 'Impose Tariff', 'No
Retaliate'))
payoff_nt_nr = (0, 0)
```

```
payoff_nt_r = (0, -5)

val_matrix = np.array([[sum(payoff_t_r), sum(payoff_t_nr)], [sum(payoff_nt_r), sum(payoff_nt_nr)]])
text_matrix = np.array([[f"{payoff_t_r}", f"{payoff_t_nr}"], [f"{payoff_nt_r}", f"{payoff_nt_nr}"]])

plt.figure(figsize=(10, 8))
sns.heatmap(val_matrix, annot=text_matrix, fmt='', cmap='coolwarm', vmin=-60, vmax=5)
plt.savefig(os.path.join(OUTPUT_DIR, "Figure_12_Payoff_Matrix_Final.png"), dpi=300)
plt.close()

try:
    df_fred = web.DataReader(['INDPRO', 'PAYEMS', 'IMPGS', 'BOPGSTB'], 'fred', '2015-01-01', '2024-12-31').ffill()
    scaler_idx = MinMaxScaler(feature_range=(0.01, 1))
    df_norm = pd.DataFrame(scaler_idx.fit_transform(df_fred), index=df_fred.index, columns=df_fred.columns)
    df_processed = df_norm.copy()
    df_processed['IMPGS'] = 1 - df_norm['IMPGS']


    def get_entropy_weights(data):
        P = data / data.sum(axis=0)
        E = - (1 / np.log(len(data))) * (P * np.log(P)).sum(axis=0)
        D = 1 - E
        return D / D.sum()


    w = get_entropy_weights(df_processed)
    df_fred['Reshoring_Index'] = (df_processed * w).sum(axis=1) * 100

    plt.figure(figsize=(16, 8))
    plt.plot(df_fred.index, df_fred['Reshoring_Index'], label='Reshoring Index')
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_13_Reshoring_Index_Fixed_Final.png"), dpi=300)
    plt.close()

    df_contrib = pd.DataFrame(index=df_processed.index)
    for col in df_processed.columns:
        df_contrib[col] = df_processed[col] * w[col]

    plt.figure(figsize=(18, 10))
    plt.stackplot(df_contrib.index, df_contrib.T, labels=df_contrib.columns)
```

```
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_14_Index_Decomposition_Final.png"),
dpi=300)
    plt.close()
except:
    pass

try:
    df_high = pd.read_csv(os.path.join(MACRO_DIR, "Chip_High_IC.csv"), index_col=0,
parse_dates=True)
    df_low = pd.read_csv(os.path.join(MACRO_DIR, "Chip_Low_Discrete.csv"), index_col=0,
parse_dates=True)
    df_prod = pd.read_csv(os.path.join(MACRO_DIR, "Chip_Production.csv"), index_col=0,
parse_dates=True)
    df_chips = pd.concat([df_high, df_low, df_prod], axis=1).dropna()
    df_index = df_chips / df_chips.iloc[0] * 100

    fig, ax1 = plt.subplots(figsize=(12, 7))
    ax1.plot(df_index.index, df_index.iloc[:, 0], color='red', label='High End')
    ax1.plot(df_index.index, df_index.iloc[:, 1], color='blue', linestyle='--', label='Low End')
    ax2 = ax1.twinx()
    ax2.plot(df_index.index, df_index.iloc[:, 2], color='black', alpha=0.4, label='Capacity')
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_16_Real_Chip_Analysis.png"), dpi=300)
    plt.close()
except:
    pass

#
==========================================================================
========
# Final Validation & Visualization (Regime Shift & Sankey & VAR)
#
==========================================================================
========

if 'history_data' in locals():
    pre_war = history_data.loc[2015:2017]
    trade_war = history_data.loc[2018:2024]

    res_pre = minimize(objective, initial_guess, args=(np.arange(len(pre_war)), pre_war.values),
bounds=bnds,
                       method='L-BFGS-B')
    res_war = minimize(objective, initial_guess, args=(np.arange(len(trade_war)), trade_war.val-
ues), bounds=bnds,
                       method='L-BFGS-B')
```

```
    labels_2017 = ["USA", "Brazil", "Argentina", "China"]
    labels_2018 = labels_2017
    colors = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728']

    fig_sankey = make_subplots(rows=1, cols=2, specs=[[{'type': 'domain'}, {'type': 'domain'}]])
    if 2017 in history_data.index and 2018 in history_data.index:
        d17 = history_data.loc[2017]
        d18 = history_data.loc[2018]
        fig_sankey.add_trace(go.Sankey(node=dict(label=labels_2017, color=colors),
                                       link=dict(source=[0, 1, 2], target=[3, 3, 3],
value=d17.values)), 1, 1)
        fig_sankey.add_trace(go.Sankey(node=dict(label=labels_2018, color=colors),
                                       link=dict(source=[0, 1, 2], target=[3, 3, 3],
value=d18.values)), 1, 2)
        fig_sankey.write_html(os.path.join(OUTPUT_DIR, "Figure_17_Sankey_Dia-
gram.html"))

if 'df_gdp' in locals() and 'df_cpi' in locals() and 'df_tariffs' in locals():
    df_macro = pd.merge(df_tariffs[['Year', 'General_Tau']], annual_gdp, on='Year')
    df_macro = pd.merge(df_macro, pd.read_csv(os.path.join(MACRO_DIR,
"US_CPI_2015_2025.csv")).assign(
        Year=lambda x: pd.to_datetime(x.iloc[:, 0]).dt.year).groupby('Year').mean().reset_in-
dex(), on='Year').set_index(
        'Year')

    df_pct = df_macro.pct_change().dropna()
    model_var = VAR(df_pct)
    res_var = model_var.fit(1)
    irf = res_var.irf(5)

    fig, axes = plt.subplots(2, 1, figsize=(10, 12), sharex=True)
    irf.plot(orth=False, impulse='General_Tau', response='US_GDP', ax=axes[0])
    irf.plot(orth=False, impulse='General_Tau', response=df_pct.columns[-1], ax=axes[1])  #
CPI
    plt.savefig(os.path.join(OUTPUT_DIR, "Figure_18_Macro_IRF.png"), dpi=300)
    plt.close()
```